DIGITAL WORKOUT: DEVELOPMENT OF RESPONSIVE WEB-BASED FOR GYM MEMBERSHIP AND MONITORING

A Capstone Project Presented to the Faculty, College of Computer Studies
Laguna State Polytechnic University
Los Baños Campus
Los Baños, Laguna

In partial fulfillment of the requirement for the degree
Bachelor of Science in Information Technology
Specialized in Service Management Program

NICA ELA D. PARAGUYA
JHON RIO GONZALES
RODELIO RAQUEL

JONALYN JOY B. LABAYNE

June 2026

APPROVAL SHEET

The Capstone Project entitled "**DIGITAL WORKOUT: DEVELOPMENT OF RESPONSIVE WEB-BASED FOR GYM MEMBERSHIP AND MONITORING"** prepared and submitted by **Nica Ela D. Paraguya, Rodelio Raquel** and **Jhon Rio Gonzales** in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology specialized in Service Management Program, is hereby recommended for approval and acceptance.

JONALYN JOY LABAYNE
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of _____.

MIKE PHILIP T. RAMOS
Member

SHERWIN B. SAPIN, EdD
Member

JOCELYN O. PADALLAN
Member

JEFFERSON L. LERIOS, DIT
Member

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology specialized in Service Management Program.

LOYD S. ECHALAR
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, EdD.
Chairperson, Research and Development Office

_____
Date Signed

RESEARCH CONTRIBUTION NO.	

ACKNOWLEDGMENT

This capstone project would not have been possible without the help, supervision and support of those people who extended their hands in completion of this paperwork. Our deepest appreciation is given to the following individuals.

To the University President, Dr. Mario R. Briones, and the Campus Director, Atty. Rushid Jay S. Sancon, for their great leadership.

To our Research Coordinator, Dr. Jefferson L. Lerios, to our dearest Dean, Dr. Loyd S. Echalar and the members of the approval committee, namely, Dr. Sherwin B. Sapin, Mr. Mike Philip T. Ramos, and Ms. Jocelyn O. Padallan for guiding us throughout the journey of this project. Their guidance and encouragement help us to keep on drawing out our ideas.

To our adviser, Ms. Jonalyn Joy B. Labayne, who shared her knowledge that really helped us to finish this study. His patience and willingness to share his expertise in programming systems, especially in revisions were needed.

And foremost, to Almighty God, who gives us wisdom and strength that guarantee the success of this capstone project. The one who truly provided everything that we needed, from financial to emotional.

THE RESEARCHERS

DEDICATION

This research paper is sincerely dedicated to the parents who served as inspiration and motivation, without their endless love and encouragement we would never have been able to complete this thesis.

To their parents,

Mr. and Mrs. Paraguya

Mr. and Mrs. Gonzales

Mr. and Mrs. Raquel

In the completion of this capstone form the beginning up to the end. And to their mentors for their mentorship throughout their project. The mentor's insights have been invaluable in the creation of their capstone project. To their friends, classmates, and siblings who cheered them up in times of hardships and difficulties.

N.D.P

J.R.G

R.V.R

ABSTRACT

A gym plays an important role in improving both physical and emotional health. However, individuals who work during weekdays often have limited time to visit a gym for inquiries, registration, and progress monitoring. This capstone project aims to develop a smooth, accessible, and timely web-based system for gym administrators, members, and prospective members. Providing accessible and accurate gym information is a primary focus of the study. The developed website offers convenient online registration, real-time monitoring of members' physical progress, accessible payment methods, and efficient membership management. The system can be accessed anytime and anywhere using mobile devices, ensuring ease of use and flexibility. Separate login modules were created for administrators and members, allowing administrators to manage and monitor member records, while members can track their workout progress, payment dues, and other relevant information in real time. Overall, the system enhances convenience, efficiency, and accessibility in gym management and membership monitoring. The system was developed using a structured software development approach and evaluated using the ISO 25010 software quality model. Evaluation results showed excellent to good ratings across key quality characteristics, including Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, and Maintainability, indicating that the system meets the required quality standards and user expectations. The findings demonstrate that the developed Digital Workout enhances operational efficiency, improves accessibility, and supports effective decision-making in gym management. This study concludes that implementing a digital platform is a practical and effective solution for modernizing gym operations and improving overall service quality.

Keywords: Gym Management, Workout Monitoring, ISO 25010, TAM, Convenience, Accessibility.

TABLE OF CONTENTS

Preliminaries	***Page***
Title Page	i
Approval Sheet	ii
Acknowledgement	iii
Dedication	iv
Abstract	v
Table of Contents	vi
List of Figures	viii
List of Tables	ix
Definition of Terms	x

CHAPTER

I	**INTRODUCTION**	
	Project Context	1
	Project Objectives	3
	Project Purpose	4
	Scope and Limitations	5
II	**REVIEWS OF RELATED LITERATURE AND STUDIES**	
	Related Literature	8
	Related Studies	9
	Synthesis and Relevance of the Study	28
III	**METHODOLOGY**	
	Project Design	34
	System Architecture	35
	System Flow Chart	35
	Use Case Diagram	37
	Project Development	38
	Business Model	39
	Materials	40
	Software	40
	Hardware	43
	Product Backlog	44
	Sprint Planning & Sprint Backlog	46
	Project Testing and Evaluation	49
	Software Testing	49
	Unit Testing	49
	Integration Testing	50
	Acceptance Testing	51
	Evaluation Procedure	52
	ISO 25010 Software Quality Model	53
	Technology Acceptance Model (TAM)	55

Population of the Study 56
Sampling Design 57
Data Collection and instrument 57
Statistical Treatment 58
Statistical Analysis of Data 59
Ethical Consideration 60

IV RESULTS AND DISCUSSIONS

Results by Objectives 62
Result of Technology Acceptance Model (TAM) 70
Result of Unit Testing 70
Result of Integration Testing 73
Result of System Testing 75
Result of Acceptance Testing 77
ISO 25010 Software Quality Model 77

V SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

Summary 78
Conclusions 79
Recommendations 80

REFERENCES
APPENDICES

Appendix A. Communication Letters and Forms
Appendix B. System Interface Screenshots
Appendix C. Diagrams
Appendix D. Survey Questionnaire and Evaluation Tool
Appendix E. Relevant Resources
Appendix F. Plagiarism Results
Appendix G. Documentation
Appendix H. Publish Format (IEEE)
Appendix I. Curriculum Vitae

List of Figures

No.	Title	Page
1	Conceptual Framework	33
2	System Architecture	35
3	System Flowchart	36
4	Use-Case Diagram	37
5	Project Development	38
6	Business Model	40
7	Software Testing	49
8	Unit Testing	50
9	Integration Testing	50
10	System Testing	51
11	Acceptance Testing	52
12	Quality Characteristic of (ISO 25010)	53
13	Technology Acceptance Model (TAM)	55
14	Administrator Interface	62
15	Member Interface	63

List of Tables

No.	Title	Page
1	Literature Matrix	29
2	Hardware	43
3	Product Backlog (Admin)	45
4	Product Backlog (Member)	46
5	Sprint Backlog (Admin)	47
6	Sprint Backlog (Member)	48
7	Respondents of the Project	56
8	Evaluation Degree, Range and Description for TAM	60
9	Scale value to Evaluate System Quality using ISO 25010	60
10	Results of Perceived Usefulness	66
11	Results of Perceived of Use	67
12	Results of Attitude towards using the System	68
13	Results of Behavior	69
14	Summary of TAM Evaluation Result	70
15	Unit Testing for Digital Workout System	72
16	Integration Testing for Digital Workout System	74
17	Browser Compatibility Testing Result	76
18	Acceptance Testing of the System	77

DEFINITION OF TERMS

AGILE Scrum	A project methodology focused on the iterative development, where requirements and solutions evolve the through collaboration between cross-functional teams.
CSS	Cascading Style Sheets (CSS) is a stylesheet language used to describe all the presentation and layout of a document written in HTML or XML.
Digital Workout	A web-based platform designed to manage gym membership, monitor member progress, and facilitate gym operations digitally.
HTML	Refers to the explanations or descriptions of all key concepts, components, or the features used within the integrated development environment (IDE) to assist users in understanding its functionality.
IDE (Integrated Development Environment)	A software platform used by developers to write, test, and debug code efficiently during the system development process.
Javascript	Scripting or programming language that allows you to write implement complex features on web pages.
Laravel 9	Develop the backend, handling server-side logic, routing, and user authentication. Its features, like Eloquent ORM, simplify database interactions, support efficient API creation, and help structure the application in a maintainable way.
Progressive Web Application (PWA)	A type of application software delivered through the web, built using common web technologies, that behaves like a native app on devices and can be accessed via a web browser without installation.
Technology Acceptance Model	A framework for assessing users' acceptance of technology, based on perceived usefulness and ease of use.

User Interface	The layout and design through the which users are interact with the system, including elements like buttons, forms, and notifications.
Visual Studio	Refers to the explanations or descriptions of all key concepts, components, or the features used within the integrated development environment (IDE) to assist users in understanding its functionality.
Web-Base System	A software application accessible via a web browser, typically hosted on remote servers, allowing users to access the system from any location with internet connectivity.

CHAPTER 1

INTRODUCTION

Project Context

The twenty-first century advanced, technology began to emerge, making many tasks easier due to the programs or systems it created. The employment of computers for information and communication has become instinctive. This is among the most important technological progressions and one of the most advantageous innovations. Computers significantly assist in the growth and enhancement of businesses and companies. Innovations like systems programs exemplify advancements that simplify the storage and protection of vital data.

Gyms have evolved into a crucial aspect of human life, offering the finest training and bodybuilding amenities within a community. Consequently, management activities are necessary to keep records of all individuals, including trainers, trainees, and employees. However, keeping data on paper is very time-consuming, so a computerized system that addresses all issues is essential. The Gym Management System is a computerized adaptation of the manual system, offering consumers unmatched data security through technology. The gym management system provides an easy-to-use solution for managing gym memberships. It aids in maintaining records of members' membership plans and facilitates communication between the owner and members. The feature-rich gym management system aids in managing and developing the club. Jeffrey’s Gym, located on the 2nd floor of Kalaw Building, Lopez Avenue corner National Highway, Los Baños, Laguna, operates as a fitness center offering

various training and exercise services. Despite its growth and increasing number of clients, the gym continues to rely on manual processes in managing its daily operations. All transactions, including membership registration, package tracking, instructor assignments, and payment recording, are done through handwritten entries in physical logbooks. This traditional method has led to several operational challenges, such as data loss, inaccurate records, and time-consuming processes that hinder the delivery of quality service to its members. As the demand for efficient service grows, the need for an automated system has become increasingly apparent.

The primary problem faced by Jeffrey's Gym is the lack of an accurate and organized system to store and manage essential records. Manual recording exposes the gym to various risks, including misplaced data, delayed retrieval of member information, and frequent scheduling conflicts. Monitoring the availability of instructors, managing the inventory of gym equipment, and tracking membership expiration dates are also complicated without a centralized system. Furthermore, payment handling through manual logs increases the likelihood of accounting errors and revenue discrepancies. These issues not only create inefficiencies for gym employees but also negatively affect customer satisfaction and trust.

The researchers proposed developing a Gym Management System specifically tailored to the operational needs of Jeffrey's Gym in order to address these challenges. The system aims to automate the scheduling of clients, assignment of instructors, tracking of members' attendance, and notification of membership renewals. It will also streamline the management of equipment

inventories, store trainers' schedules, and centralize all payment transactions into a secured and accessible database. By providing real-time information and automated alerts, the system will significantly reduce the administrative burden on the staff, eliminate manual errors, and ensure that operations run smoothly and efficiently.

Through the implementation of the Gym Management System, Jeffrey's Gym will be able to modernize its operations, improve overall service quality, and enhance the customer experience. The system will allow for faster, more reliable transactions and better management of gym resources. It will also enable the management to generate useful reports, supporting data-driven decision-making that can contribute to the growth and sustainability of the business. Ultimately, this project seeks to provide Jeffrey's Gym with a robust technological solution to meet the demands of its expanding clientele and continuously evolving fitness industry standards.

Project Objectives

This capstone project aims to create a web-based gym membership management system for Jeffrey's Gym that would enhance business operations, improve customer convenience, and strengthen financial control. In particular, this project has the following aims:

1. To create an accessible and responsive web-based platform that provides a seamless user experience across all devices (desktop, tablet, and mobile) without the need for native app installation.

2. To implement core functional modules for user accounts, membership management, payment processing, and progress monitoring.
3. To evaluate the acceptability of the developed system using ISO 25010 and Technology Acceptance Model (TAM).

Project Purpose

This study aims to create a web-based system for managing gym memberships that will streamline business processes and produce detailed payment reports. The objective of this system is to foster healthier lifestyles by improving the accessibility and organization of fitness services. It directly contributes to the Sustainable Development Goals (SDGs), especially SDG 3: Good Health and Well-being, by promoting regular physical activity. SDG 9: Industry, Innovation, and Infrastructure via digital solutions in the fitness sector, and SDG 12: Responsible Consumption and Production through diminished dependence on paperwork and encouragement of sustainable management practices. This project will benefit the following:

Gym Owners. This study will help the gym owner as this will help improve gym operations by automating and streamlining processes such as member registration, class scheduling, and payment management. This system will reduce the reliance on manual tasks, which will make gym management more efficient and allows owners to focus on enhancing customer service and their business.

Gym Administrator/Staff. The system will reduce the workload of gym administrator/staff by automating routine tasks such as record, and scheduling

management. This allows staff to be more productive and focus on delivering better customer service and maintaining gym facilities.

Gym Members/Clients. Members will benefit from a more organized and accessible experience. They will be able to register online, book classes, track attendance, monitor their fitness progress, and receive timely notifications about schedules or promotions, which will lead to increased engagement and satisfaction.

Trainers/Coach. Instructors will be able to manage their class schedules, view participant lists, and track individual member progress more efficiently. This enables better workout planning and personalized training for clients.

Future Researchers. This study will help future researchers by providing a basis for exploring advancements in digital health management systems, fitness technology, and business process automation. It can serve as a reference for further studies in improving the user experience and operational effectiveness of fitness centers.

Scope and Limitation of the Study

This study aims to develop a web-based gym membership system designed to enhance the business processes of Jeffry's gym, situated on the 2nd floor of the Kalaw Building at the intersection of Lopez Avenue and National Highway in Los Baños, Laguna. The online system aims to simplify membership processes by enabling users to register, renew, and manage their memberships through the

web. It also encompasses functionalities like training plans, reservations for classes, and monitoring of individual advancement. Tools are provided to gym staff for the efficient management of member records, attendance monitoring, and payment processing. Moreover, the platform fosters member involvement via tailored exercise regimens and automated alerts. The web system comprises various modules and features.

Login Module. The Login Module allows users to securely access the web-based gym system using their registered credentials. It ensures that only authorized members, trainers, and administrators can enter the platform. This module also includes authentication features to protect user data and maintain account privacy.

Account Management Module the Account Module manages user profiles, allowing members to update personal details such as name, contact number, and fitness goals. It provides account settings for users to modify passwords and manage their login preference. Administrators can also view and control user access levels to maintain proper system management.

Equipment Module the Equipment Module helps monitor gym equipment availability and condition through a web-based interface. It allows staff to record maintenance schedules, track usage, and update inventory status. This ensures that all equipment is functional and ready for members' use at any time.

Trainer Module the Trainers Module manages trainer profiles, schedules, and availability in the system. It allows trainers to update their personal information,

assigned clients, and training sessions. Gym administrators can easily assign members to trainers and monitor performance or attendance.

Attendance Module the Attendance Module records the check-in and check-out of both members and trainers. It provides an automated log of attendance history, which helps in monitoring participation and discipline. The module also supports viewing daily, weekly, or monthly attendance reports for better management.

Payment Module the Payment Module allows members to settle their membership fees conveniently through GCash or cash payments. It keeps a record of all transactions, including payment dates, amounts, and methods used. The system automatically updates membership status once payment is confirmed, ensuring smooth and accurate tracking.

Report Module the Report Module generates detailed reports related to membership, attendance, and payments. It provides visual summaries and downloadable reports to help gym administrators make data-driven decisions.

CHAPTER 2

THEORETICAL FRAMEWORK OF THE PROJECT

Review of Related Literature, Studies, and Systems

This chapter presents a comprehensive review of related literature and studies, both local and foreign, found in reliable sources, including online articles, blogs, and other trustworthy published materials. It will serve as the researchers' guide in conducting the study. Additionally, it connects and examines several earlier studies that will help in explaining and understanding the significance of the study.

Gym Management System

The Gym Management System (GMS) is a modern digital solution aimed at automating and optimizing the operational processes of gyms. This paper explores the design, implementation, and benefits of the GMS, focusing on enhancing efficiency, accuracy, and overall user experience. By integrating modern web technologies, the GMS streamlines membership management, class scheduling, and payment processing, addressing the shortcomings of traditional manual systems. The system also offers advanced reporting and analytics, empowering gym owners to make data-driven decisions for better business management. (Chokhandre et. Al, 2025).

Gym Management System is an online service that can be setup for your gym to help manage classes, memberships, receive payments (merchant and cash), keep track with detailed statistics, customer management, surveys and it

even has an online store so you can sell products to your customers. Manage your trainers and class schedules. Customers can see their own statistics and payment history. Complete Online Store for your products, membership plans & passes. (Karmakar, 2022) The Gymnasium Management System is a collection of wellness centers designed to make dealings with clients easier and more efficient. The project is a computerized computer-based framework and is used to deal with gym members. This study expects to build a framework for people who want to maintain their health and physical wellbeing on a regular basis, relying on the gym. The manager can see all members from the gymnasium and also their details. The plan framework can supervise recreation center members, gymnasium instructors, trainers, gymnasium gear, diet of gymnasium members, instalments of gym members, instalment of instructors, and equipment details. The framework, which will store information, will check the legality of data provided by members, stores data of members as per their ID, and creates reports for various IDs. This framework is secure, and to login to the framework one's username and password is required.

This Automated Gymnasium Management System is more reasonable than the manual information base, since it gives trainers automated framework support, huge capacity limit, more faithfulness and high safety. (Lakshitha and Sirisuriya, 2021). The study was carried out based on a group of indicators that allowed to identify the elements that should be considered essentials in a GMS. To study the relevance of the use of GMS in the district of Leiria, a survey was carried out with the owners and/or managers of gyms in the area. The study

also served to contrast the study on the desired characteristics for a GMS and the perception that gym staff have about them. Another of the elements included in the research work are the study of the main methodologies, tools, and technologies used in the implementation of a GMS. The system must be a solution that includes a web application and a mobile application. These technologies include Microsoft Azure, Terraform, Microsoft ASP NET Core and React. Scrum was used as a software development methodology. The core concept of this architecture is the concept of Microservice. Microservices is an architectural style that structures an application as a collection of services. The idea is to group the businesses that look similar in the system and develop them as individual units. The development of a microservices system becomes to develop small system's units that are more manageable, maintainable, and scalable (Delgado, 2021). Gym Management System allows the user to store the food details, employee details, the details of person who is in the gym, gym equipment details etc. This software package allows storing the details of all the data related to a gymnasium. The newly developed site for Gymnasium is more suited than the manual database because it provides the facilities like, large storage capacity, high speed, more accuracy and high security (Vazhachariccka, 2020).

According to Suryawanshi et al. (2024), a gym management system is designed for members and gym owners to easily manage its everyday transactions, increase member satisfaction and business growth. Most of the gym's transactions were recorded manually which is challenging and time-consuming. Large number have been encountered over years through manual

labor. It is also hard for the management to identify the mistakes once it is done manually. People nowadays prefer an online process as it is more convenient. A web-based system management lessens the burden on both parties of members and the management. An easy way to track every detail of the members, including membership, payment and their preferences.

According to Nivedita et al. (2025), the full-stack Web-Based Fitness Trainer and Client Interaction System which is designed to bridge the gap between trainers and clients. This platform assures trainers to provide customized workout plans, monitor their client's progress, and provide real-time feedback while earning outside gyms. Clients can log their workout sessions in order to keep track of their fitness journey while can also follow remote training sessions. The system ensures efficiency, engagement, and accountability, making it a more reliable platform for both trainers as well as clients. This paper details the implementation, features, as well as advantages of this platform.

According to Ahmed & Nayeem (2017), any business that does not have a website is missing out on one of the most powerful marketing tools available to them. The main reason that it is important for businesses to have a website is how people are likely to find you. These days most people will go online and research products and companies before they make a purchase. For this reason of dependency on online, we develop a website for a gymnasium in which customer can get all information about the gymnasium and he/she can access or enrol at the gym via online. Usually, the client uses MS Excel or paper, and maintains their records, however it is not possible them to share the data from multiple system in

multi user environment, there is lot of duplicate work, and chance of mistake. When the records are changed, they need to update each and every excel file. The Smart Gym Management System eliminates most of the limitations of the existing software. Increasing efficiency and effectiveness, automation, accuracy, user-friendly interface, information availability, communication capacity, maintenance, cost reduction makes our system smarter than the existing system. We integrate some new and prominent features along with all the necessary features.

The COVID-19 pandemic has disrupted numerous industries, with gyms being particularly affected. Their operational setup, involving multiple touchpoints, poses an increased risk of virus transmission. Consequently, there is a pressing need to minimize human interaction within gyms, especially for tasks like attendance tracking and registration. To address this concern, a proposed solution involves implementing a contactless attendance monitoring system Integrated with an Enterprise Resource Planning (ERP) platform. (Shan, 2023). With the development of China's economy, more and more people spare no effort to participate in fitness activities and join in the gym exercise. In order to adapt to the rapidly developing social market and meet the needs of the public, the gym management system with comprehensive functions is essential for both operators and participants. Design a gym management system with practical application value, aiming to carry out the comprehensive management of all kinds of data and information generated in the daily business of the gym, and assist users' work (Zhao et al., 2023).

Hardware et al. (2025), stressed that web-based gym management systems simplify the process of membership management as well as workout tracking and scheduling sessions for users. The system uses automated functions with real-time data combined with secure payment features as well as other methods to enhance operations and customer interaction. The system provides individualized workout plans together with automatic reminder systems which delivers an improved fitness journey to customers. The goal is to apply Industry 4.0 components and principles to a gym environment with the aim to create a smart gym. The project approaches the problem in six steps. The first step is to conduct a literature review to gain knowledge and insight into work relating to Industry 4.0, the fitness industry and the future of gyms. Step two looks at applying the industry 4.0 design principles to a gym environment. In Step three we develop a smart gym model and also define a smart gym (since no definition currently exists). In Step four we design two concept model systems and show how to collect relevant data for gym owners and managers. In Step five we implement the two test systems at the Van der Stel gym in Stellenbosch. We do this to prove the concept systems in practice and show that it is possible to extract data and information from gym equipment. Step six lists the results and conveys the conclusions. The project provides evidence that it is possible to extract usable data and information from gym equipment (Kotzé, 2017).

Gym is a best place in order to help people to do exercise since there is a lot of equipment provided. This equipment could assist people do the exercise easily and no need to spend a big space to make the movements. Busy people

nowadays decided to do exercise at the gym because at the gym they can have a trainer or instructor who will guide them to do exercise correctly and accordingly. The trainers who have more knowledge about health and expert in running the equipment at the gym will make the customers will feel confident to be trained by the trainer instead of doing exercise by their own without using any equipment such as doing physical exercise at the home. People nowadays also chose gym for their place to do exercise because of time constraint. They do not have so much time to do regular exercise because of other commitments in life including they are busy with their carrier and other work in their daily life (Ismail, 2016). Gym is equipped with all the modern machineries and other supportive items to provide a better service to its customers. so the numbers of members are tent to increase day by day and Gym Management has decided to go for an IT solution in order to handle the increased volumes. Software is developed with the aim of replacing the current manual system which was using at the Gym.

The system supposed to provide easy access of the resources of the gym to its members while empowering the management process of the gym. Following major functionalities such as member management, attendance management, charges management, maintenance management and reports generation were identified. Access control was achieved with the username and password. User access levels were also implemented to control the accessibility of the system. The system has the capability of suggesting the best workout schedules that suit to the members while keeping the harm full workouts away depending on the members' physical condition. The system also capable of

notifying the delayed payments and pending services of exercise machines. High level of customization is embedded in to the system where administrator can control the access and appearance of the system. (Gamage. K.G.S.T, 2017).

The gym has all the equipment which are mostly up to date according to current fitness trends and also has other helpful products to give its clients a better service. As a result, the number of members would likely grow daily, leading the gym management to choose an IT solution to handle the rising traffic. Software was created and developed using an incremental and iterative development process based on the requirements. The manual system now in use at the gym is intended to be replaced by the JK Fitness Office software. The system's goal is to give members quick access to the gym's resources while enhancing the management of the facility. The primary capabilities that were discovered were as follows: member administration, attendance management, payments and expense management, and report generating. It was possible to control access using the login and password. Additionally, user access levels were implemented to control the system's accessibility. The system is capable of tracking members whose packages or memberships are about to expire and will immediately email them a membership reminder. The system can be highly customized, allowing the administrator to manage user access and how the system looks. (Nishantan, M. 2023) Nahaga Gym Sabu Seba previously relied on manual systems that hindered administrative processes such as member registration, payment verification, and membership tracking. This study aims to develop a web-based information system to automate and streamline these processes. Using the Agile development

method, the system was designed with key features including online member registration, package selection, digital payments, downloadable membership cards, and real-time administrative dashboards for monitoring members and revenue. (Achyar Fadil Alboneh Alboneh & Snae M. 2025).

An analysis of the web application supporting the management of the gym was carried out, which was designed in terms of the theory of control and systems. The obtained results of measurements of input and output quantities of selected subsystems were used to carry out the identification, as a result of which models of web application subsystems were obtained. On the basis of the obtained subsystem models, appropriate models were then designed in Simulink for the purposes of simulation and comparative studies of the behavior of subsystem models in relation to real systems that are web applications, including an appropriate analysis of the web application system model was carried out under the name of the gym on its selected subsystems, including in particular the subsystem called Generating a diet.(Jerzy Tchórzewski et. Al 2023).

Gym is becoming the most favorite place to exercise physic. The most common reason go to the gym is fills self-confidence up to the brim with building muscles throughout the body, but most did not last long then stop doing workout. Unfamiliar with the equipment, poor knowledge about training program and diet program are the main problems. In another case android is the world’s most popular mobile platform providing Android Studio where someone can setup a new Android project for different types of Android apps within seconds. (ST Bana Handaga. 2016).

The study of (Arolkar S. et al. 2022). This app will show the gyms nearby to the user along with the joining fees. This will help the user to compare and decide which gym they want to register at. It will keep a record of the attendance of the user. The user can scan a machine in the gym using this app and it will show them the right way to use the machine. The main focus of this project is to build a dedicated app for gym requirements and managing all the data without the hassle of the manual work. One of the important features of this app is the use of augmented reality to identify the equipment and display the exercises to be performed on this equipment.

The GMS-BPI Gym System is like a super cool computer program for gyms! It helps gym owners and people who go to the gym. For the owners, it makes things easier like signing up new members, planning classes, fixing gym stuff, and keeping track of money. There’s even a fancy screen that shows how the gym is doing. But the awesome part is how it helps people who work out. GMS-BPI uses special gadgets to collect info about how fit they are. So, people can see how healthy they’re getting, make goals, and get workout plans just for them. And it figures out a special number called the Body Performance Index to show how fit someone is overall. (A. Srivastava, S. Annamalai. 2024).

The Smart Gym Management System is to enhance and upgrade the existing system by increasing its efficiency and effectiveness. The software improves the working methods by replacing the existing manual system with the computer-based system. The Smart Gym Management System automates each and every activity of the manual system and increases its throughput. Thus, the response

time of the system is very less and it works very fast. It provides a quick response with very accurate information regarding the users etc. The transaction reports of the system can be retired as and when required. Thus, there is no delay in the availability of any information, whatever needed, can be captured very quickly and easily, with the reduced cost of maintenance (Kumar et al., 2020).

Virtual Gym Management for a gymnasium in which user can enroll to the system via online and he can access his workout or trainer and diet or consultancy at the best time and best place for their schedule. For user interaction with trainer or consultancy, system has provided with video chat & text chat. After specific period i.e. when certain amount of data gets enter into database, system will automatically recommend user for workout and diet by giving him report through his profile. User have to give feedback about gym and about its working as it will be helpful for management team to modify system if required (Jathar et al., 2018).

Currently, in a modern world where people are getting busier, it is hard for them to take time to work-out or exercise regularly by themselves. People have been using the gym as a place to make their body fit and it is right as Health and Immunity are an important part of a person’s life and everyone would always like to be fit and healthy. To achieve that level requires motivation and discipline. And lack of motivation results in quitting the gym in a very short span. The slightest changes and every other important aspect can be hard to keep track of them altogether, now using smart gym approach, the slightest change can be measured, using IoT and later one can curate all the training regiments, diets and exercises properly according to it. We propose an overall IoT-based-system to monitor the user’s

Health and Fitness Records in an effective way of using a database management system. It plans to collect data from the machines when the user uses it, keep track of its workouts and diet intake, with a gym social media that would help to maintain a competitive environment, also including management of membership, payment, trainers, and employees. This would result in a whole new way of looking towards the gym.(Krish Bhanushali. 2022).

Optimization of Gym Check In And Membership Information System QR Code Scanner Based

The development of information technology provides opportunities to improve operational efficiency in various fields, including membership management at gyms. A QR code scanner-based information system is an effective solution for managing gym member check in. Users can check in simply by scanning a QR code that is linked to their membership data. This system not only facilitates the attendance process but also helps managers monitor the number of member check in, remaining attendance quota, and attendance time. This research aims to design and implement a QR code scanner-based check in system at the gym, as well as analyze the impact of the system on operational efficiency. (Badriah N. & Shandy S. V.).

Gym Check system is a mobile application that will allow gym users and management to see the amount of overall traffic and individual equipment availability in their gym. The Gym Check system combines usage data, machine-tracking technology, and check-in information to provide a user-friendly interface

for both gym-goers and gym management. This system provides real time tracking of individual pieces of equipment so that gymgoers don’t have to deal with unavailability and waiting, and gym management can manage demand for each piece of equipment. (Kazmi M. et al. 2016).

According to (Villanueva M.C. 2025). There is a growing trend of people who join fitness clubs to improve their health and physical appearance. Olympics Fitness Gym is one of the well-known gyms in Tacloban City. Despite being famous, it still uses manual monitoring and data management schemes in running the fitness club. The proposed QR Code Technology thus seeks to automate the manual process of recording sales, generating reports, monitoring the attendance and subscriptions of the customers by increasing its efficiency and effectiveness. The researchers integrated a new QR Code technology to improve the attendance and membership subscription monitoring features to bridge these practical gaps. As a result of the study, the researchers suggest automating the manual process and creating a management system with integrated QR-Based technology to increase its efficiency in attendance and sales monitoring as well as in data management A gym is a favourite place for physical exercise. The most common reasons for going to the gym are build self-confidence by building muscles throughout the body, but mostly did not last long then stopped doing the exercise. Unfamiliar with equipment, Poor knowledge of training programs and diet programs is a major problem. Online platform is the world’s currently most popular platform for commercials.(Muizzuddin Muzaini. 2022).

Gym memberships are offered to the public for a fee to provide cardiovascular and strength conditioning equipment that promotes exercising behavior and improves health. A health insurance company in Reno, Nevada offers the Motivate U wellness program, which includes gym membership and a personal trainer, at no cost to selected insured members to promote healthy lifestyle changes; with an anticipated return-on-investment (ROI) due to decreased healthcare costs and medication expenses associated with the treatment of comorbidities of obesity. The purpose of this prospective study was to evaluate the health outcomes and cost-benefit ratio of Hometown Health Insurance's Motivate U Program. This study compared Motivate U participants' cost of current medications and number of health care visits, along with changes in body mass index (BMI), blood pressure, and laboratory values at baseline and after completion of an 8-week gym membership program. A decrease in healthcare expenses would demonstrate an economic benefit for Hometown Health Insurance that would be significant enough to offer free gym membership coverage for all their members (Orange, 2017). As organisations grow and diversity, traditional manual record keeping and event management methods become impractical and error-prone. A membership management system offers a comprehensive and automated solution to handle membership-related tasks effectively. This paper aims to develop a system that provides an intuitive and user-friendly experience for administrators and members, enabling them to manage their profiles seamlessly, access personalised services, and participate in various activities. Also, it can streamline administrative processes, automate manual tasks, and

centralise data to enhance association efficiency. By simplifying these aspects, associations can focus on strategic initiatives and foster more robust engagement within their community (Hambali et al. 2024).

Vizta Gym is a fitness center with a modern, clean, comfortable concept and complete facilities. Vizta Gym, which is present in several places in busy centers or malls in Medan City, also makes it easier for its members to carry out physical fitness sports activities wherever the Vizta Gym branch is located. However, the member verification process is still an obstacle. Someone who activates a member at one Vizta Gym branch and wants to do sports activities at another Vizta Gym branch must be verified manually. Apart from that, members often forget the sports schedule that has been prepared or planned in advance and the member's active deadline. The aim of this research is to build a Vizta Gym membership data processing system using the prototype method and unified modeling language (Sembiring, 2023)

Gym Scheduler: A Gym Equipment Reservation System

In the study of Ebrahimi et al. (2017), the solution was to give a user the ability to reserve different types of equipment for a specific period. This will help reduce the wait times and can help one stay on schedule. The reservations have a maximum limit to equipment as well as the time they are allocated to the student. Gym Scheduler will prove to be really helpful in such circumstances. With this tool, you never have to worry about the availability of any of the machines. With as much as three clicks, you can schedule any instrument of your choice for a limited

time. Web-Based Information System for Managing Gymnasium Membership have been developed to meet the needs of both gym owner and members. This project report outlines the design, development, and implementation of this innovative web-based information system, which aims to revolutionize the gym management experience. The Web-Based Information System for Managing Gymnasium Membership is a comprehensive software system that leverages modern web technologies to streamline gym operations, enhance member engagement, and provide data-driven insights for informed decision-making (Siagian, 2023).

Jain (2015), propose a framework for a smart gym. The framework consists of a simple model and prototypes which can be implemented in a gym. The framework is basically an implementation of the smart city based on IOT. The foremost objective of the proposed framework is to develop an intrusive technology that will not only provide valuable and innovative services to the user by integrating the user and the smart devices but also provide an efficient utilization of the resources present in the gym. With the growing interest in fitness and easy access to fitness information, the demand for fitness is rising. In order to attract longer-term subscriptions, many large chain fitness clubs typically employ a strategy where the longer the membership contract, the lower the average monthly fee. However, this could potentially result in problems when individuals wish to cancel or modify their plans later. Because of the high cancellation fees, some individuals choose to transfer or sell their memberships on social media platforms like

Facebook, PTT, and Dcard (Taiwan internet forum). However, such transactions often lead to disputes and inconvenience (Tsai et al., .2025).

Gimify: Development of a Progressive Web App to Reserve Gym Assets

Over the years, technology has revolutionized our world. Technology has created amazing tools and resources, putting useful information at our fingertips. Over the past few years we have gotten accustomed to use services with a computer; terms like online banking and online shopping have become part of our daily life. More and more businesses have gained popularity by making very easy to buy goods and services; companies like Uber, Airbnb, Amazon, Trivago, Booking and many more have built their fortune on this concept. Some of these companies allow people to book hotels, taxies, flights, trains, restaurants without any human interaction whatsoever, increasing revenue for the company and making the process so much easier for the end user. (G. Giunchi, P. di Milano, 2018)

Enhancing Gym Attendance and Engagement Through a User-Centered and Co-Design Approach for Key Features in a Mobile Application

This thesis explores a user-centered approach to enhancing gym attendance and engagement by addressing key motivational barriers identified through in-depth research. Guided by service design principles (Stickdorn et al., 2011), the development of a mobile application emerged as a strategic solution based on insights from co-design sessions, participant feedback, and expert interviews. These interactions underscored the convenience and effectiveness of a digital tool

in providing consistent motivational support tailored to individual behaviors. (Narges Pesian. 2025)

Fitness Center: An Automated System for Gym Notification with Client Attendance and Guidance System

There are many people who are not able to use it due to various reasons. So, we are making an Android app which will help them. This project will allow the gym owner to manage all the receipts and also notify the users about their fees, mark their attendance by own, provide them proper diet plans as per the BMI and body type. Also this application will help them by providing exercise guidance clips. Gyms and workout studios often only focus on getting people to purchase year-long memberships. It is very difficult for both the members and the trainer to keep all the paper receipts safe. Also it is difficult to manage all the client at a time and mark their attendance and provide exercises and diet plans. (Dadge R. et al. 2022)

Virtual gym management: Web based application to sell fitness products and equipment

In this paper our blueprint is to create a smooth web user interface and it can be run smoothly in windows or android devices. The web application will provide new updated calisthenics goods and users can buy using their credit or debit cards without hassle or hurdles in transaction. Gym Website, as described above, can lead to error free, secure, reliable and fast management system. It can assist the user to concentrate on their other activities products rather to concentrate on the

record keeping. Thus it will help users to better utilization of resources in websites. The user credentials records saved in database without redundant entries. There will be a good communication between user and admin panel regarding goods quality and quantities and we will make sure to satisfy customers in every aspect. (P. N. Reddy et al. 2022).

The use of IOTA's technology to create a decentralized, fee-free system for tracking gym attendance and entry could be beneficial and an advantage to the gym as well as its members. Removing the inefficiencies and security threats linked to traditional methods can create an easy, safe, and resistant solution to record gym members' check-ins and check-outs in real-time. The goal is to transform the gym administration process with an automatic and decentralized solution that improves operations efficiently and enhances user experience. This decentralized system for tracking gym attendance intends to eliminate many of the current problems and difficulties faced by gym operators and members by utilizing IOTA's Tangle technology to provide a more effective and safer alternative for managing gym attendance. (Phung L. 2025).

Swolemate: A Fitness Assistance Smartphone Application with Gym Search, Workout, Meal Plan, Physical, and Mental Status Checking Feature

Physical and mental health status both contribute to a person's overall well-being. A person's physical well-being can be satisfied by engaging in adequate fitness training, eating healthy foods, getting enough sleep, and maintaining a disease-free body. Existing research on fitness assistance systems did not

disclose a modern ICT technology-based online solution for examining a person's physical and mental health status by examining their fitness score, exercise, body mass index (BMI), behaviors, and disease-related data.

The existing articles did not exhibit online solutions for calorie and BMI score examination, client prevalence-based GYM search, workout planning, calorie-based meal planning, daily task scheduling, health specialist search, and client fitness-related inquiry assistance. To unravel the previous challenges, this article dispatches a fitness assistance smartphone application that issues client priority-based GYM search, physical and mental health status checking, calorie-based meal planning, workout progress tracking, BMI and calorie consumption tracking, daily task scheduling, law enforcement assistance, health specialist search, and fitness-related inquiry assistance features. (M. Chowdhury, N. Hadi. 2024).

Online Payment Transaction System

Online payment transaction system is playing a crucial role in the economic development of countries. It indicates the growth in revenue generation and reduction of black money. This chapter aims to know the genesis of online payment transaction system using plastic money card. The development and enhancement in security for online payment transaction system due to the rise in plastic money card fraud with change in innovative and sophisticated fraudulent patterns of cybercriminals, fraudsters, or hackers is the reason behind the study. This chapter presents the growth of the online payment transaction system from start to present

and the drivers behind its development, including digital technological development (Khattri et al., 2024).

In the study of Bakar and Said (2019), the customer able to book for classes and personal trainer via online. In addition, manager is provided instant access to sales data thus able to make decision faster. OGBS is proposed as a solution to current way of inefficient booking management that requires the stakeholders to update class availability and perform booking task in a manual way. The system was developed by adapting the Waterfall Model of SDLC. Online Membership Business While the work of Shakoor et al., (2018) presents a design for online database and transaction management. The design system can manage employee, member, facilities, payroll, receipts, and products information. It also provides the facility of search & advanced search for searching the records efficiently & immediately.

Synthesis and Relevance of the Study

The reviewed literature consistently highlights the growing need for automated, web-based solutions in gym management due to the limitations of manual systems. Traditional record-keeping methods are prone to errors, inefficiencies, and difficulties in data retrieval and sharing, especially in multi-user environments. As a response, modern Gym Management Systems (GMS) integrate core functionalities such as membership tracking, payment processing, class scheduling, and report generation.

Several studies underscore the value of real-time features like online booking, automated reminders, personalized workout plans, and remote trainer-client communication to enhance user experience and operational efficiency. Moreover, microservice-based architectures and the use of agile methodologies like Scrum have been noted for improving scalability and maintainability in GMS development.

Security and data integrity are also common themes, with systems incorporating login authentication and secure payment features (Lakshitha & Sirisuriya, 2021; Shan, 2023). The COVID-19 pandemic further accelerated the shift to digital platforms, emphasizing the need for contactless and remote management solutions (Zhao et al., 2023; Shan, 2023).

Importantly, studies also reveal the broader impacts of GMS, such as improving health outcomes and reducing healthcare costs through programs integrated with gym memberships. Advanced features like analytics dashboards and real-time monitoring not only support members but also empower gym owners to make informed decisions.

Literature Matrix

Table 1. Literature Matrix

Author(s)	Year	Title	Findings
Bakar N. & Said M.	2019	The Development of an Online Gym Booking System (OGBS) using	Online reservations increase organizational effectiveness and client

		Online Membership Business Model	convenience using the Waterfall Model (SDLC)
Chokhandre S. et al.	2025	Gym Management System	The automation streamlines membership management, class scheduling, and payment processing, using the Agile methodology.
Delgado H.	R. 2021	Gym-Suite an Extensible Platform for Gyms Management	Microservice architecture enhances the manageability, scalability, and maintainability of systems. Scrum-based software development; microservices-based technological design was utilized.
Ebrahimi et al.	A. 2017	Gym Scheduler: A Gym Equipment Reservation System	The u se of WebStorm, Heroku, and Node.js in a collaborative project context typically implies an Agile or incremental development process reduces wait times for equipment and raises user satisfaction.
Hambali al.	et 2024	Design and Evaluation of a Web-based Membership Management System	Automated systems improve member satisfaction and operational efficiency.

Jain A.	2015	A smart gym framework: Theoretical Approach	Smart gym technology can provide innovative, user-centered services, that focus on the creation and integration models between users and smart devices.
Kendiga J.	2017	GYM Management System Of Taurama Fitness Center	Object-oriented approach with UML diagrams and Prototype Mode shows that the manual systems negatively affect customer satisfaction and overall service efficiency.
Sembiring R.	2023	Design Of a Web-Based Membership Data Processing System at Vizta Gym Using a Prototype Method	Web-based system streamlined member verification, improved schedule reminders, and tracked membership activity.
Shakoor M. et al.	2018	Database and Transactions Management System for a Smart Gym: LAYYAH FITNESS CENTER	A centralized database enhances efficiency and data handling.
Siagian H.	2023	Web-Based Information System for Managing Gymnasium Membership	The system enhances the experience of members through better service access

Suryawanshi A. et al.	2024	An Analysis of the Gym Management System Study	Web-based are more efficient and reduce errors.
Vaishnavi G. R. et al.	2024	The Gym Management Website	Improved gym operations through smart features like wearables, community forums, and automation
Zhao D. et al.	2023	Design and Implementation of Gym Management System	JSP enables dynamic interaction between users and the system

Conceptual Framework

Figure 1 illustrates the development process of the web-based gym management system. The creation of the system began with gathering information from the gym owner and staff, which provided insight into existing operational challenges and the specific needs of the users. This initial step guided the planning and design of the system's structure. The system is composed of four main modules: the Account Module, the Membership Module, and the Payment and Report Module. Each module plays a specific role in automating and streamlining gym operations, and the framework highlights how these components interact to create an integrated platform.

The development process used essential technologies such as a web-based frontend, backend server logic, a relational database, and an integrated development environment (IDE) for efficient code management. To organize tasks and monitor progress, the team adopted the Agile Scrum methodology, breaking

the development into short iterations with regular feedback and continuous improvement. To assess user experience and acceptance of the system, the Technology Acceptance Model (TAM) was applied. Furthermore, ISO 25010 quality standards were used to evaluate key software attributes such as functionality, usability, efficiency, and reliability to ensure the system meets the required quality benchmarks.

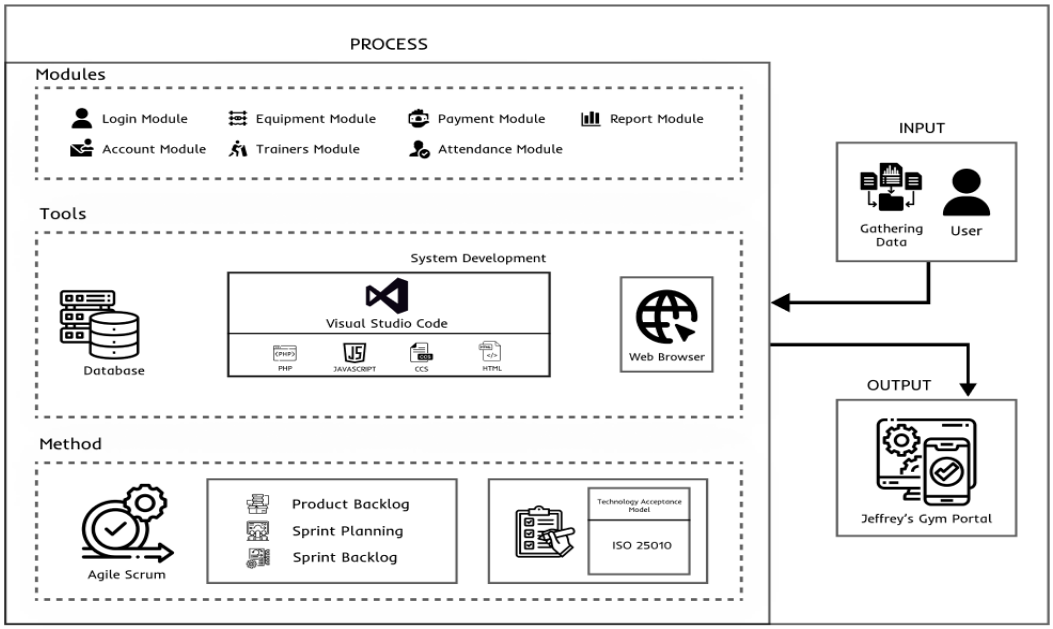


Figure 1. Conceptual Framework

CHAPTER 3

METHODOLOGY

This chapter includes the project design, project development, and software and hardware requirements for developing the system.

Project Design

This chapter applied architecture principles and practices to develop the system architecture that provides insight into the linkages between end-user requirements.

The system architecture shown in Figure 2 below includes a login feature that allows both gym members and staff (such as trainers and administrators) to set up accounts and enjoy a personalized fitness experience. By using logins, users can access all features within the system securely and efficiently. Members are encouraged to create accounts to access features such as personal profiles, membership details, and class bookings. Likewise, staff members can create accounts to manage member information, oversee payment processes, and generate activity reports. The Account Module handles core user-related functions, including login, registration, profile management, and logout. This module ensures that only authorized individuals can access system features. It provides each user with a unique profile that stores personal information, access roles, and activity history. The Membership Module is designed to manage gym subscriptions. Members can register for new membership plans, renew their existing plans, or request upgrades. Administrative staff can approve new memberships, monitor subscription status, and maintain records of active and expired accounts. This

module ensures streamlined membership processing and tracking. The Payment Module manages all financial transactions within the system. Members can securely process their subscription payments, whether one-time or recurring. Staff members can track completed and pending payments, send payment reminders, and generate digital receipts. This module enhances financial transparency and simplifies transaction handling. The Report Module provides essential tools for data analysis and decision-making. Through graphs and visual reports, users—especially staff and administrators—can view metrics such as membership growth, payment trends, and attendance rates. This information allows for better planning, improved member engagement strategies, and effective operational management.

System Architecture

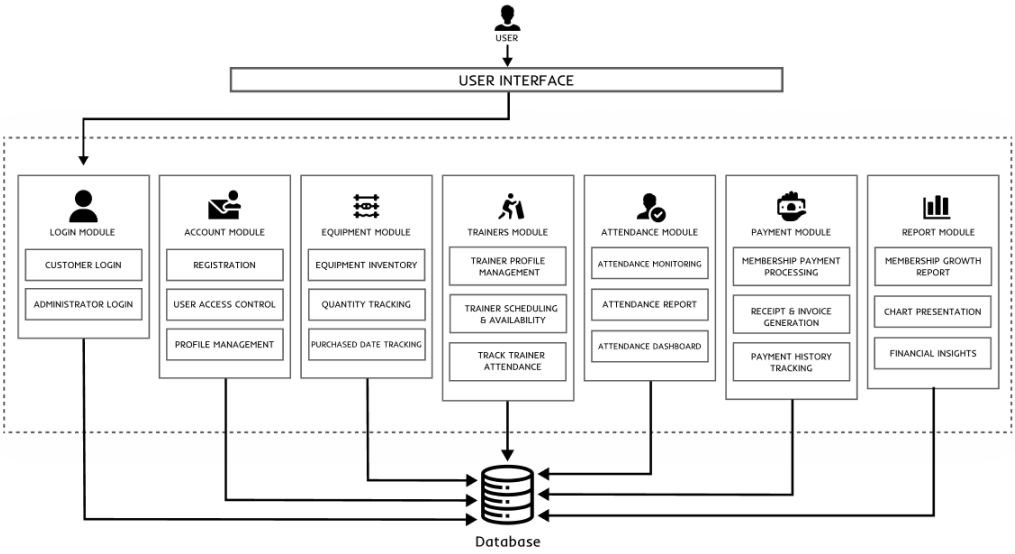


Figure 2. System Architecture

System Flowchart

Figure 3 illustrates the interaction between the Gym Management System and its three primary users: Admin, and User. After logging in, users are directed

based on their roles. Regular Users can perform actions such as viewing notifications, sending enquiries, and requesting personal trainers. These interactions are stored within the Gym Management System. Admins and Staff, on the other hand, have extended functionalities including managing member registrations, enquiries, and attendance; handling staff management tasks such as monitoring staff attendance and managing accounts; generating reports; and managing notifications and messaging. Both users, staff, and admins can log out after completing their activities. All operations performed by admins and staff directly interact with the system's central data repository, ensuring that all updates are properly maintained. The flowchart visually highlights the distinct but interconnected functionalities available to each user type, supporting efficient management and smooth user interaction within the system.

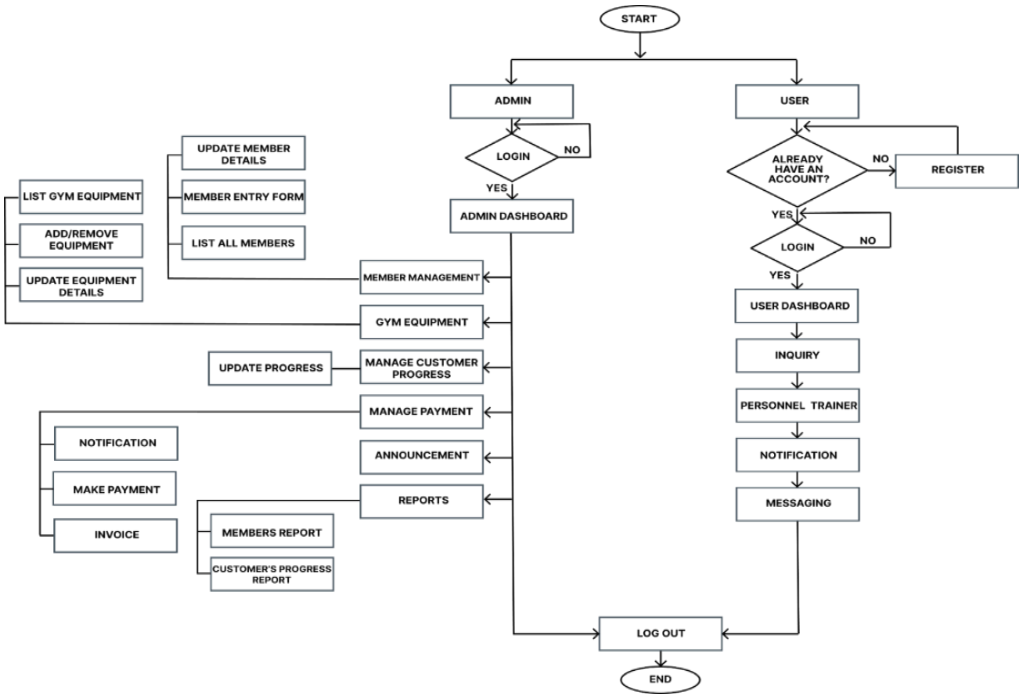


Figure 3. System Flowchart

Use Case Diagram

Figure 4 illustrates the use case diagram that provides a visual representation of the different functionalities available in a gym management system. The Administrator is responsible for overseeing and managing the entire system. This role has access to a wide range of functionalities, including members management, where the administrator can add, update, or remove gym members; manage gym equipment, which involves tracking and maintaining gym equipment; and manage customers progress, where individual user fitness progress can be monitored. Additionally, the administrator can handle payment tasks related to billing and fees, creation of announcements for system-wide notifications, staff management for adding and removing staff employees, and report panel for generating usage reports and payment records. Meanwhile, members interact with the system mainly for personal account activities. Both users can register, log in, and logout, this suggests that users can sign up, enter their credentials to access their profiles, and securely exit the system.

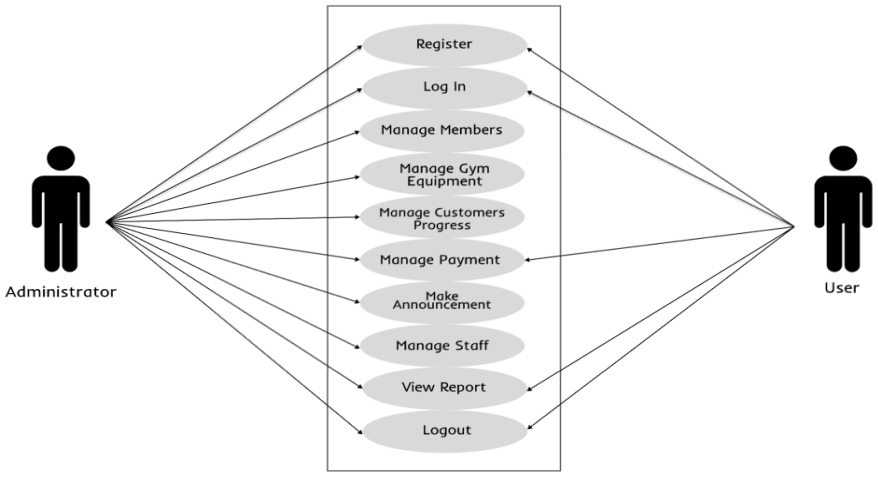


Figure 4. Use Case Diagram

Project Development

Figure 5 illustrates the use of the Agile software development life cycle in designing the web-based system. The team adopted Agile Scrum, breaking the project into phases with 1–4-week iterations. This approach emphasized regular collaboration with users and continuous improvement. Key practices included managing a product backlog, sprint planning, daily scrums, and delivering a finished product at the end of each iteration.

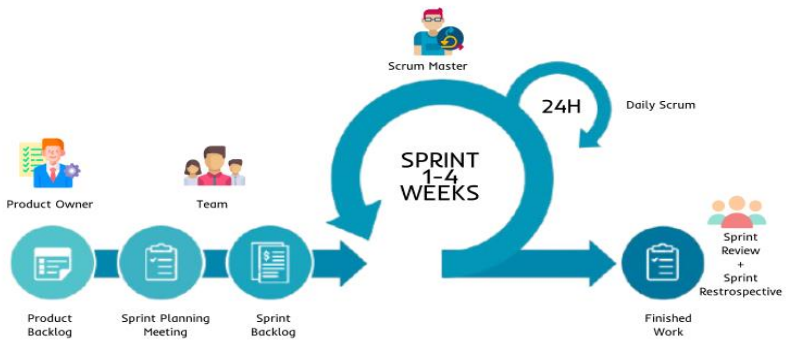


Figure 5. Agile Scrum

The Gym Management System is designed to support the needs of gym administrators, trainers, and members by providing a centralized platform for managing day-to-day gym operations. The system enables administrators to oversee essential components such as member registration, attendance monitoring, trainer availability, equipment inventory, announcements, financial records, and overall gym performance. Members are included within the system's scope by allowing administrators to track their membership plans, attendance, and payment status. Trainers may also be part of the system, enabling administrators to view active trainers and manage their assignments.

Design

The system is designed to replace the traditional manual handling of gym memberships, attendance logs, and training schedules. A clear and intuitive storyboard was created to guide the development process, outlining the structure of the modules, features, and interface components. The design emphasizes a user-friendly experience that allows members to easily navigate the system, access gym services, and manage their personal fitness information. On the administrative side, the design supports efficient data management, quick access to membership records, streamlined payment processing, and effective communication with members and trainers. This approach ensures that the system enhances operational efficiency, improves tracking of gym activities, and strengthens coordination among members, trainers, and administrators.

Business Model

The Business Model as shown in Figure 5 for the Web-Based Gym Membership and Monitoring System highlights how the platform delivers value to both gym members and administrators. Its core value purpose is to simplify and automate gym application processes while ensuring secure data management and providing real-time updates. The system operates with key partners such as Jeffrey’s Gym and relies on essential resources including a skilled development team, hosting services, and development tools. Key activities involve platform design and development, database maintenance, automation implementation, user training, and system testing following TAM and ISO 25010 standards. The primary customers are gym members, with administrators and gym owners serving

as secondary users. Customer relationships are strengthened through accessible feedback, automated updates, notifications, and developer support. The system is delivered through the gym's website or portal and supported by on-site client orientation. The cost structure focuses on development and maintenance, continuous system improvement, and hosting services needed to keep the platform operational and efficient.

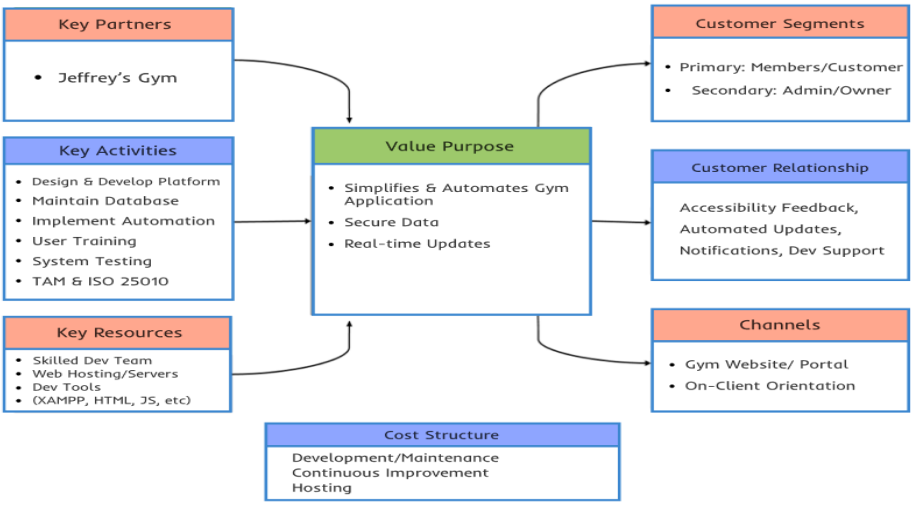


Figure 6. Business Model

Materials for Software Development

This section shows the software and hardware equipment that used in developing the project.

Software

Listed below is the software to be used by the developers in developing the web-based system including the reasons and features.

XAMPP, the developers can mimic an online server environment on their local computers, making sure that all aspects of the system are thoroughly tested and optimized prior to actual deployment onto a live server. This arrangement significantly helps in quicker development, simpler debugging, and more effective management of database-driven functionalities.

PhpMyAdmin is used by developers to manage and deploy the website's database. It allows them to store and organize critical data, such as product details, user information, and transactions. Developers rely on phpMyAdmin to ensure the database is updated in real-time, keeping the website synchronized with the latest information.

Visual Studio Code is the system's core code writing, editing, and management application, helping developers to build the website, interface with the database, and implement features like community feedback and virtual rewards. The project is coded using HTML, CSS, and JavaScript on the front end and PHP and SQL on the back end. Visual Studio Code also contains helpful testing and debugging extensions. phpMyAdmin, on the other hand, is utilized for database management, holding significant information like user profiles, feedback, sentiment results, reward points, milestones, and redemption histories. It enables developers to update records, run queries, back up data, and help the system's reliability and smooth functioning.

HTML (Hypertext Markup Language) is used by developers to build the structure and layout of a website. It provides the foundation for all content, such as

text, images, and navigation, allowing developers to organize and present information in a way that is both functional and user-friendly.

MySQL serves as the relational database for the application, storing and organizing all essential data such as user accounts, posts, messages, and other transactional records. It ensures data integrity, allows complex queries, and supports efficient data retrieval, which is crucial for the application's performance. With MySQL, the application can handle multiple users and large volumes of data effectively while maintaining security and consistency.

Cascading Style Sheet (CSS) is used to control the visual appearance of the system, ensuring a neat, modern, and user-friendly experience. It defines the layouts, colour schemes, fonts, spacing, and other stylistic features that make the interface attractive and simple to navigate. This plays a crucial role in maintaining the cleanliness, modernity, and user-friendliness of the system for both user and administrators.

Laravel 9 is used to develop the backend of the application, providing a robust framework for handling server-side logic, routing, and user authentication. It simplifies the management of requests and responses, allows for efficient creation of APIs, and helps structure the application in a modular and maintainable way. Additionally, Laravel's built-in features like Eloquent ORM make database interactions smooth and secure, reducing development time.

JavaScript is used to add interactivity and dynamic features to the web-based system. It enhances the user experience by allowing real-time updates, form

validations, interactive charts for community sentiment analysis, and responsive elements without reloading the entire page. JavaScript also helps in managing user actions, such as submitting feedback and claiming digital rewards, making the system more engaging, efficient, and user-friendly.

Hardware

Table 2 shows the basic equipment needed for the project to run smoothly. You need at least 4GB of RAM for the software used. The use a quad-core CPU because the software is heavy-duty. It's best to have Windows 10. The used one to two computers to develop the web-based system.

Table 2. Hardware Specification

HARDWARE	SPECIFICATION
RAM	8GB
CPU	Intel(R) Core(TM) i5-4210U
OS	WINDOW 10

Table 3 shows the basic specifications required for PCs and laptops to ensure the system runs smoothly. Due to potential compatibility issues with newer operating systems, it is recommended to use Windows 10. Additionally, a web browser and a stable internet connection are necessary to fully access and utilize the system's features.

Table 3. Minimum Requirements for PC or Laptop

MINIMUM REQUIREMENTS FOR PC OR LAPTOP	SPECIFICATION
RAM	4GB
HARD DISK	At least 2GB free space
PROCESSOR	Intel Core i5-4210U
OS	WINDOW 10

Table 4 shows the minimum specifications required for a smartphone or tablet to operate the designed system smoothly and without errors,as mentioned. To ensure full system functionality, a stable internet connection and a compatible web browser are also necessary.

Table 4. Minimum Requirements for Smartphone or Tablet

MINIMUM REQUIREMENTS FOR SMARTPHONES OR TABLET	SPECIFICATION
ANDROID	Android 10
RAM	4GB
PROCESSOR	Quad-core
BROWSER	At least 2GB free space

Product Backlog

Table 5 shows the Product Backlog for the admin section, outlining key tasks essential for the system's development. These tasks reflect the core responsibilities of administrators, including logging in, managing user accounts and permissions, handling orders placed by members, accessing analytics

reports, and updating gym services and membership plans. Each item is accompanied by an estimated time frame to guide development efforts. This backlog serves as a foundational reference for gathering client requirements and defining the development roadmap. By prioritizing tasks, the team can efficiently allocate resources and ensure that the system effectively supports administrative operations and decision-making.

Table 5. Product Backlog (Admin)

Priority	User as Admin Story	Est. hr/day
1	As admin, I want to log in to the system	2hrs
2	As admin, I want to manage user accounts and permissions	3hrs
3	As admin, I want to view and manage orders placed by members	4hrs
4	As admin, I want to access analytics reports for members 3hrs activity and payments	5hrs
5	As admin, I want to modify gym services and subscription plans	5hrs

Table 6 presents the Product Backlog specifically for the gym member side of the system. It identifies core user functionalities that enhance engagement and efficiency within the platform. Gym members are expected to perform actions such as creating accounts, browsing services and plans, managing their subscriptions, and tracking their workout and payment history. The backlog includes estimated development time for each task, helping developers prioritize user-centered

features that streamline gym service interaction. This structured approach ensures a user-friendly and efficient experience that aligns with the project's goal of promoting active fitness participation and digital convenience.

Table 6: Product Backlog (Member)

Priority	User as Member Story	Est. hr/day
1	As a member, I want to create an account to access the system	2hrs
2	As a member, I want to browse available gym services and membership plans	3hrs
3	As a member, I want to add preferred services	2hrs
4	As a member, I want to securely complete payments through the checkout process	3hrs
5	As a member, I want to view my activity and payment history	2hrs

Sprint Planning

Table 7 presents the Sprint Backlog for the admin panel of *Digital Workout*, highlighting key features that developers must prioritize during system development. These tasks are derived from the product backlog and refined into actionable development items. Each task is mapped with estimated working hours and aligned with the four phases of the sprint process: Planning, Coding, UI Design, and Testing/Debugging. During the planning phase, the development team identifies how each feature will be implemented. The coding phase involves the integration of required functionalities. UI design ensures features are accessible and intuitive, while testing/debugging guarantees system reliability and

error resolution. For the admin side, tasks include implementing secure login, managing staff and customer accounts, monitoring payments and analytics, and modifying gym services and plans. These features ensure efficient control over operations and data visibility for gym administrators.

Table 7: Sprint Backlog (Admin)

Priority	Story	Task	Est. hr/day
1	As admin, I want to log in to the system	Design and develop a secure login interface with authentication.	2hrs
2	As admin, I want to manage user accounts and permission	Develop a management interface to add, update, or delete customer and staff accounts.	3hrs
3	As admin, I want to view and manage member subscriptions and payment records	Build a dashboard for tracking payment history, subscriptions, and financial status.	3hrs
4	As admin, I want to access analytics reports on gym usage and revenue	Design an analytics module to visualize member engagement, usage stats, and total earnings.	3hrs
5	As admin, I want to update gym services and subscription packages	Create a content management module for editing available services and pricing.	2hrs

Table 8 presents the Sprint Backlog for gym members (user side), detailing user-focused features to enhance engagement and streamline access to services

in the *Digital Workout* platform. Features include registration, browsing workout plans or services, selecting and purchasing plans, and tracking progress or payment history. Each user story has been converted into a specific development task, with estimated durations.

Table 8: Sprint Backlog (Member)

Priority	Story	Task	Est. hr/day
1	As a member, I want to create an account to access gym features	Design and implement a registration and login form with validation.	2hrs
2	As a member, I want to browse available gym services and workout plans	Develop a services catalog showing available programs with details and pricing.	3hrs
3	As a member, I want to add preferred plans to my subscription	Implement a selection module for choosing services or plans to subscribe to.	2hrs
4	As a member, I want to complete my membership purchase and get confirmation	Develop a checkout process with secure payment and confirmation functionality.	3hrs
5	As a member, I want to view my workout progress and payment history	Build a user dashboard displaying training logs and transaction records.	2hrs

Project Testing and Evaluation

The developers tested each part of the Gym Management System to make sure everything works properly. They checked all features such as member management, attendance, trainers, payments, and equipment to confirm that the system performs as expected. Through these tests, they ensured the system is easy to use, accurate, and ready for actual operation.

Software Testing

In Figure 7, the image outlines the stages of software testing, where each component of the software is examined to confirm it functions properly. It demonstrates how various parts are tested to ensure they perform their intended tasks accurately.

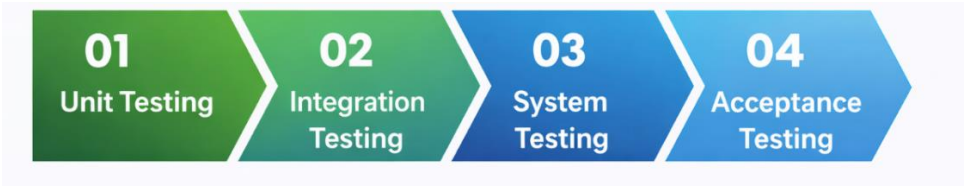


Figure 7. Software Testing

Unit Testing

As shown in Figure 8, the unit testing process involves a detailed inspection of individual components within the system. Each element is tested thoroughly to ensure it performs its intended role and aligns with the overall development objectives.

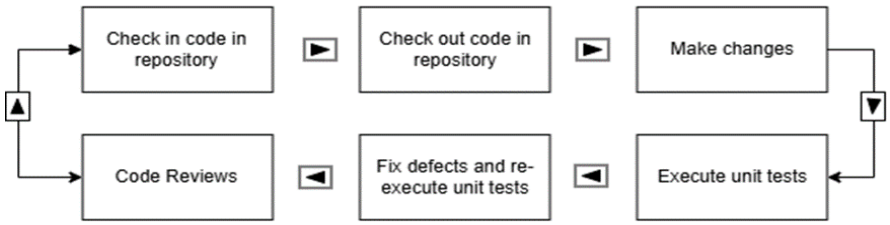


Figure 8. Unit Testing

Integration Testing

The figure 9 illustrates the Integration Testing phase, where developers verified that the modules defined in the system design work correctly together. This stage focuses on testing how individual components interact, identifying any bugs or issues, and ensuring that all parts function seamlessly as a whole.

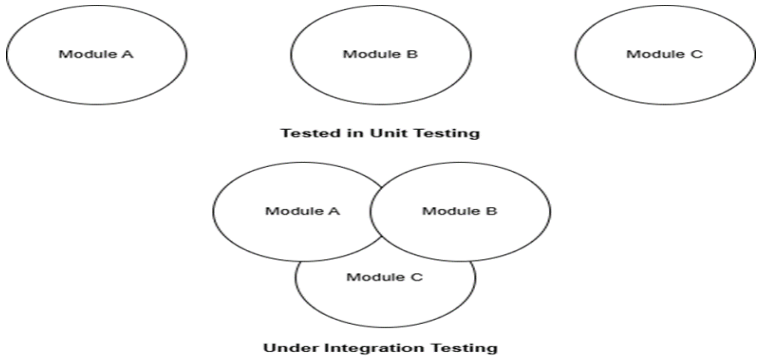


Figure 9. Integration Testing

System Testing

Figure 10 describes the compatibility testing process, where the gym management system is tested across different web browsers. This phase evaluates how consistently the system performs in various web environments to ensure that its features, user interface, and functionalities operate correctly

regardless of the browser used. System testers verify that all components of the gym management system function as intended, ensuring that users can efficiently manage member registrations, track memberships and attendance, process payments, generate reports, and receive notifications and alerts seamlessly across different browsers.

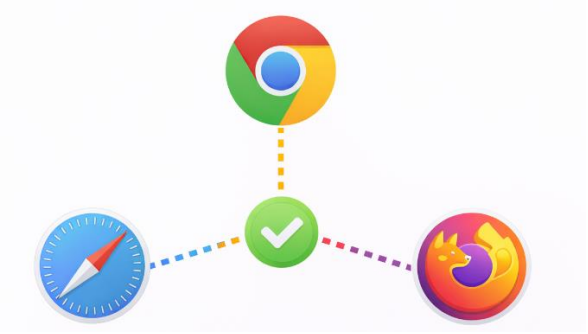


Figure 10. System Testing

Acceptance Testing

Figure 11 presents the acceptance testing phase, where the system is evaluated based on the requirements agreed upon with the user. The project is considered accepted if it satisfies the user's expectations. This stage involves a comprehensive review of the system, including its design, to ensure it meets the defined acceptance criteria. Developers conduct formal testing to confirm that the system aligns with user needs, giving the user the final decision on whether to approve it.



Figure 11. Acceptance Testing

Evaluation Matrix

Standards, like ISO 25010 to ensure the quality of the developed system, these standards provide the important elements of a high-quality product. The ISO 25010 standards describe the quality model, which is the basis of a system for evaluating the quality of a product. The ISO 25010 contains eight quality attributes that describe assessment features of a system. The IT experts evaluated the developed project by the ISO 25010 software standard to study and learn about the performance and general features of the Web-based system. The most commonly used to determine the consistency of any program is ISO 25010. Evaluation should consist of Functional and non-functional aspects, Internal and external design, and structure. To evaluate the system to be developed, the developers do the beta tests for the I.T experts. The developers used the ISO 25010 quality model as the standard for Likert scalar. Likert scalar is answerable by five responses which is five (5) - Excellent (4.20-5.0) four (4) -Good (3.40-4.19), three (3)-Fair (2.60-3.39), two (2) - Needs Improvements (1.80-2.59), and one (1)-Poor (1.0-1.79). Each module and functionality of the application would be tested

to ensure the development process goes well. This testing will be performed as the web-based system develops to detect any faults.

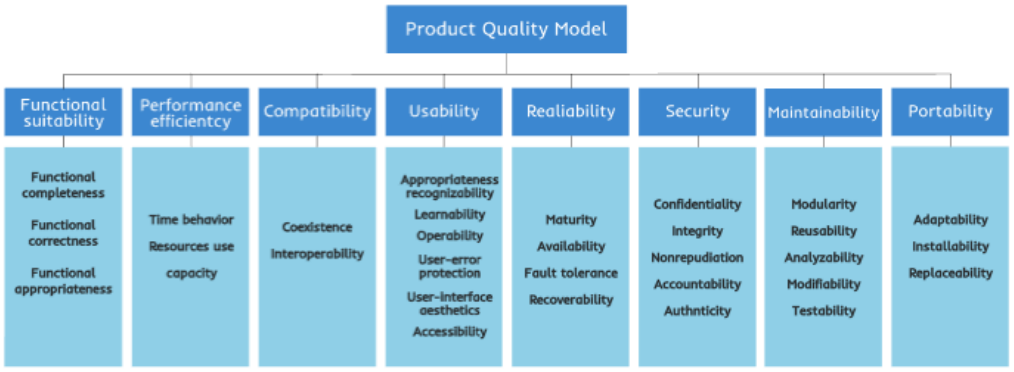


Figure 12. Quality Characteristic of ISO 25010

The respondents and IT experts evaluated the developed project by using ISO 25010 software standard to study and learn about the performance and general features of the web-based system. The most popular way to determine the consistency of any program is ISO 25010. Assessments should include both functional and non-functional aspects, as well as internal and external design and structure.

For evaluating the system to be develop, the developers do beta test for the users and IT specialists. The developers used the ISO 25010 quality model as standard for Likert Scalar. The Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability are the basis on testing the Web-based Employee Profiling Management System. Likert Scalar is answerable by four responses which is SA = Strongly Agree, A = Agree, D = Disagree and SD = Strongly Disagree.

Functional Suitability. The system was evaluated to ensure that its features meet the needs of gym administrators, trainers, and members. The developers demonstrated that the system effectively supports essential tasks such as member registration, attendance monitoring, payments, and trainer management.

Reliability. IT specialists examined the stability of the system by verifying that its features run accurately and consistently. The goal was to ensure the system remains dependable during daily operations.

Usability. The developers designed the interface to be simple, organized, and easy to navigate. Testers reviewed whether administrators and staff could easily understand and use the system without confusion.

Security. User information such as member profiles, payments, and attendance is protected through secure data handling. Testers evaluated the system's measures to prevent unauthorized access and maintain data privacy.

Compatibility. The system was checked across multiple browsers, devices, and screen sizes to confirm that it works properly regardless of the user's platform. This ensures accessibility for different types of users.

Maintainability. IT specialists assessed how easily the system can be updated, modified, or improved in the future. This evaluation ensures long-term system support and enhancement.

Flexibility. The system was examined for its ability to adapt to future changes, such as adding new modules, adjusting features, or scaling gym

operations. Testers ensured that the system structure can accommodate growth and updates without major issues.

Safety. The system was evaluated to ensure that data handling and operations do not pose risks to user information or system stability. Safety checks confirmed that processes run securely without causing errors, data loss, or operational hazards.

Technology Acceptance Model (TAM)

The Technology Acceptance Model (TAM) helps users understand and utilize the system by assessing its performance, functionality, and usefulness. TAM gathers data through surveys to evaluate system effectiveness and address user requirements, such as interface and benefits. It includes four characteristics: Perceived Ease of Use. Perceived Usefulness, Attitude Toward Using, and Behavioral Intention to Use, along with Social Influence and Information Awareness. By applying TAM in surveys, the evaluation identifies areas for improvement, predicts future user interactions, and determines user preferences. satisfaction, and behavior.

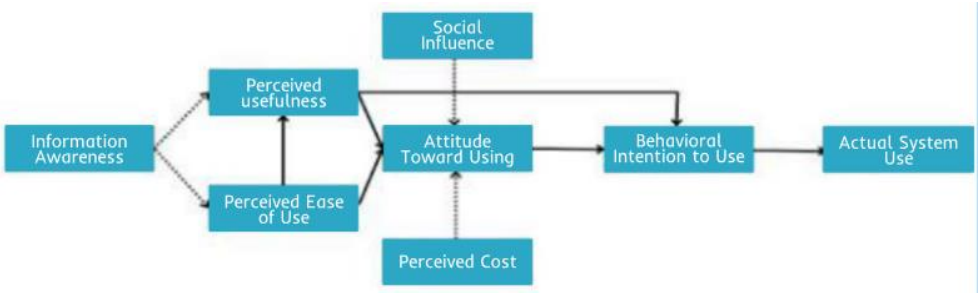


Figure 13. Technology Acceptance Model (TAM)

Population of the Study

In evaluating the system, three (3) IT expert was selected to assess and ensure the secure handling of collected data and to detect potential issues such as errors, bugs, or system failures. The research involved illustrating, analyzing, and evaluating the feasibility of developing a web-based system, focusing on the overall performance and features of the developed platform. An IT expert with knowledge of online trading platforms was consulted to provide a detailed analysis and assessment of the system's performance and reliability.

Table 9. Respondents of the Research Project (ISO 25010)

CATEGORY OF RESPONDENTS	SAMPLE
IT Expert	3
Total	3

Table 10 presents the respondents involved in the research, consisting of fifty (50) participants, divided equally between customers/gym users and admins/owners. These respondents analyzed and evaluated the system by providing insights into its effectiveness in supporting business operations and enhancing the user experience. Their feedback helped determine whether the system meets the requirements for daily activities and management tasks.

Table 10. Respondents of the Research Project (TAM)

CATEGORY OF RESPONDENTS	SAMPLE
Customer/Gym Member	48
Admin/Owner	2
Total	50

Sampling Design

Purposive sampling is a method where the analyst intentionally selects individuals from the nearby area to participate in the evaluation based on specific criteria. This type of non-probability sampling was used because it enables quick and convenient data gathering from participants who are easily accessible within a short timeframe.

Data Collection Instrument

For this study, data was gathered through interviews and structured surveys designed to support the development of Digital Workout: A Comprehensive Web-Based Solution for Gym Membership Efficiency and Engagement. The surveys served as the basis for understanding user needs, system effectiveness, and areas for improvement.

Two sets of questionnaires were utilized in this project. The first set followed the ISO 25010 standard, evaluating key quality attributes such as functional suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability. Respondents rated each item using a five-point

Likert scale with the following classifications: 5 – Excellent (4.20–5.00) - 4 – Good (3.40–4.19) - 3 – Fair (2.60–3.39) - 2 – Needs Improvement (1.80–2.59) - 1 – Poor (1.00–1.79). This survey was answered by one (1) IT expert to evaluate the system's quality and performance from a technical standpoint.

The second set of questionnaires applied the Technology Acceptance Model (TAM) to measure factors such as Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, and Behavioral Intention to Use. This questionnaire was rated on a four-point scale ranging from 1 (Not Acceptable) to 4 (Highly Acceptable), and was distributed to system end-users, including gym members and administrators.

To ensure the reliability of the survey instruments, Cronbach's Alpha testing was conducted. This statistical method is widely used to measure the internal consistency or reliability of a set of scale or test items, particularly in Likert-scale questionnaires. A higher Cronbach's Alpha value indicates greater internal consistency. In this study, Cronbach's Alpha was applied separately to both the ISO 25010-based questionnaire and the TAM-based questionnaire. A reliability coefficient of 0.70 or higher was considered acceptable, with values closer to 1.00 indicating excellent reliability. The results of the reliability test confirmed that the items in both questionnaires consistently reflected the constructs they were intended to measure, thereby validating the robustness and dependability of the data collection instruments used in evaluating the system.

Statistical Treatment

The used a percentage method to determine the respondent's level of evaluation. The developers also used like scale technique to interpret the collected data and evaluate application features. The evaluation contains a series of question that may be answered by one of four responses/impressions, which were categorized in the following order: Four (4) - Strongly Acceptable, three (3) - Acceptable ,two (2) - Not Acceptable, one (1) - Strongly Unacceptable. To eliminate partiality and misinterpretation of the data, the undefined response will be deleted.

Statistical Analysis of Data

The frequency of each response was determined by the number of respondents that answered the particular item.

The Likert Scale was utilized by the proponents as a survey technique, as shown in the table above. It was used to assess the end users' satisfaction with the system. The percentage approach was also employed by the proponents to measure the respondents' evaluation dimensions. The formula used was as follows:

$$\% = \frac{f}{n} \text{ x } 100$$

Where

f = frequency

N = Total number of Respondents

This project used the Likert Scalar method to evaluate and analyze the gather data.

For TAM:

$$x = \frac{(SA*4) + (A*3) + (D*2) + (SD*1)}{N}$$

Where

x = Mean

N = Total Number of Respondents

SA = Strongly Agree

A = Agree

D = Disagree

SD = Strongly Disagree

Table 11. Evaluation Degree, Range and Description for TAM

Degree	Range	Description
4	3.26 – 4.00	Strongly Agree
3	2.51 – 3.25	Agree
2	1.76 – 2.50	Disagree
1	1.00 – 1.75	Strongly Disagree

For ISO 25010:

$$x = \frac{(E*5) + (G*4) + (F*3) + (NI*2) + (P*1)}{N}$$

Wh

x = Mean

N = Total Number of Respondents

E = Excellent

G = Good

F = Fair

NI = Needs Improvement

P = Poor

Table 12. 5-Point Likert Scale to evaluate system quality using ISO 25010

Degree	Range	Description
5	4.21 – 5.00	Excellent
4	3.81 – 4.20	Good
3	2.61 – 3.80	Fair
2	1.81 – 2.60	Needs Improvement
1	1.00 – 1.80	Poor

Ethical Consideration

To verify the protection of the data, this paper strictly apply ethical guidelines to assure the protection and rights of all the participants involved.

The protection of the participants through the application of ethical principles is significant in any research study. First communicate to the owner of Jeffry's gym to inform them and ask to permission in conducting interview. And provides a consent letter with the signature of our research adviser to be given to our client and for us to have a permission to conduct the interview and collect the data. To ensures that the participants should understand what they will be asked during the interview. This means, to participate in a capstone project, participants need to be informed about the research and it depends on them if they are willing to participate or not. Assure that the data we collect and the information that the client said during the interview is safe and won't be disclosed to other. The question used were checked by the research adviser before it was carried out.

CHAPTER 4

RESULT AND DISCUSSION

This chapter presents the results and the discussion to support the methodology utilized in the achievement of the objectives.

Results by Objectives of the Study

1. Design and develop a responsive web-based system for gym membership and monitoring, tailored to the needs of gym administrators and members to streamline membership management, track workout progress, and improve overall gym operations.

Administrator Interface

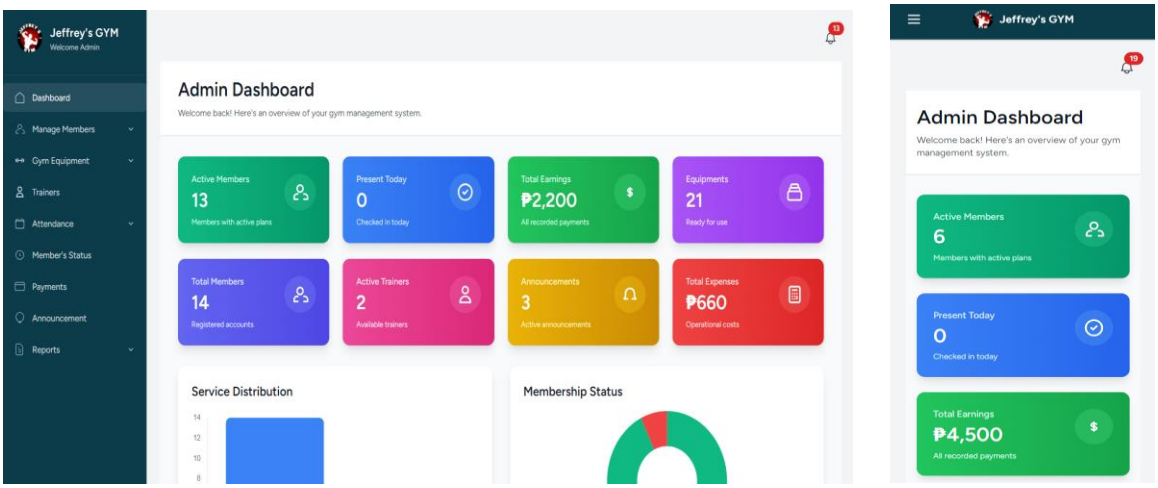


Figure 14. Dashboard Management Interface for Administrator

Figure 14 shows the admin dashboard shows the current status of the system, allowing administrators to easily monitor all activities. It provides quick visibility on whether there are new members or important updates. Through this dashboard, the admin can immediately take action and manage the system more efficiently

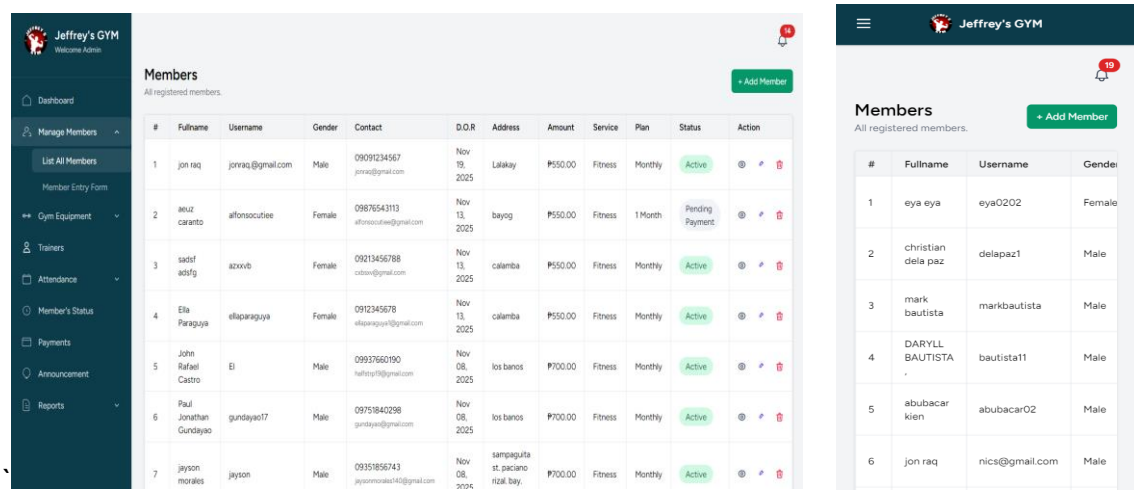
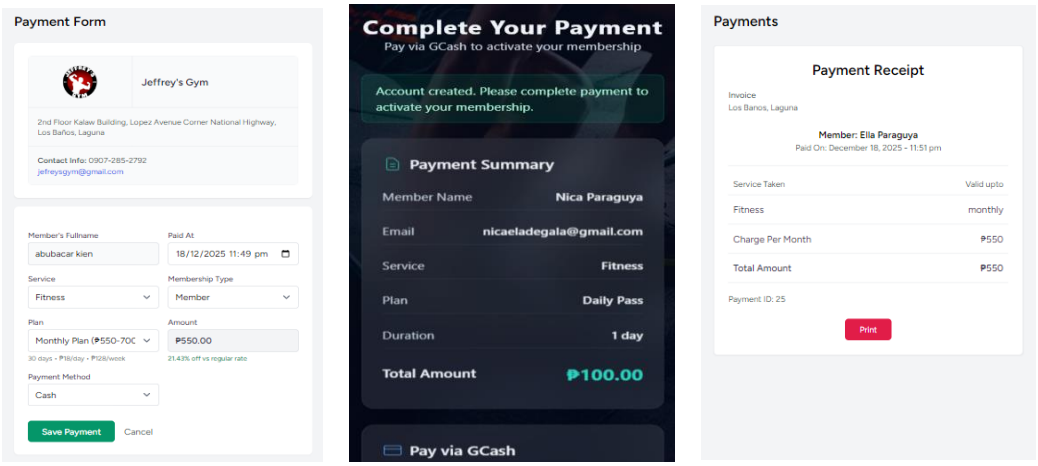


Figure 15. Member Management Interface for Administrator

Figure 15 shows the admin's ability to manage all gym members. Through the Manage Member feature, the admin can edit, update, and organize member information as needed. It also displays a complete list of registered gym members for easy monitoring and management.



(a) (b) (c)

Figure 16. Payment Option and Process Feature: (a) Cash Payment, (b) Gcash Payment, (c) Generated receipt

Figure 16 shows the payment options and workflow of the gym membership system. The system provides two payment methods: Cash Payment and GCash

Payment, allowing members to conveniently settle their membership fees. For cash transactions (a), the system displays the computed total amount, enables the admin to input the cash received, and automatically calculates the change, reducing manual computation errors. For GCash payments (b), the system displays the gym's QR code and allows the admin to input the reference number provided by the member after completing the GCash transaction. Once the payment is successfully verified, the system processes the transaction accordingly. after a successful payment through either method, the system generates a receipt (c) that contains the transaction details such as the member's name, payment method, date of transaction, and total amount paid. This generated receipt serves as official proof of payment and ensures proper documentation and monitoring of all payment transactions within the system.

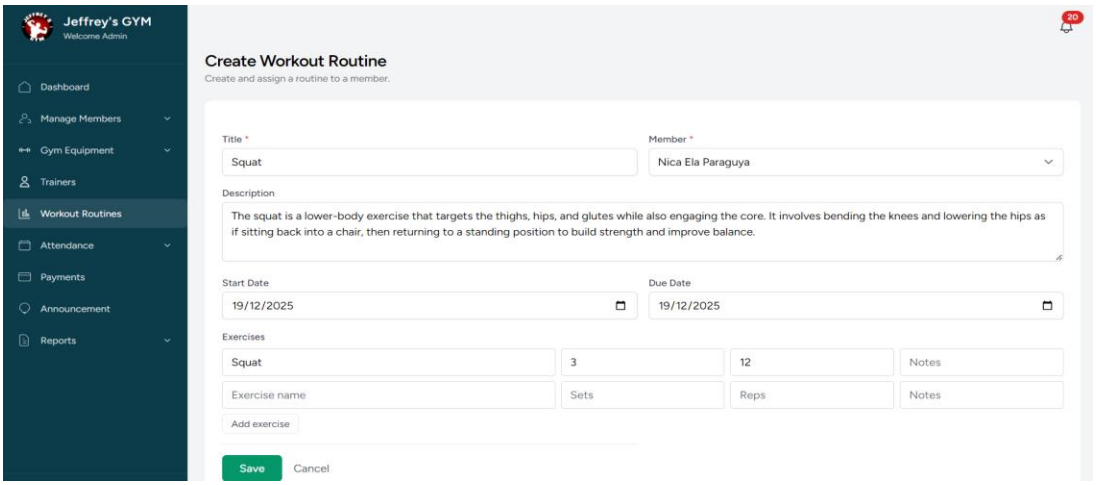


Figure 17. Administrator Interface for Progress Monitoring

Figure 17 shows the administrator interface for workout progress monitoring. Through the Create Workout Routine feature, the admin can create and assign workout routines to specific members, including exercise name, sets, repetitions, start date, and due date. This interface allows the admin to define work

out details such as squats, monitor assigned routines, and track member progress over a given period. It also supports proper organization and monitoring of workout plans to ensure members follow their prescribed fitness programs effectively.

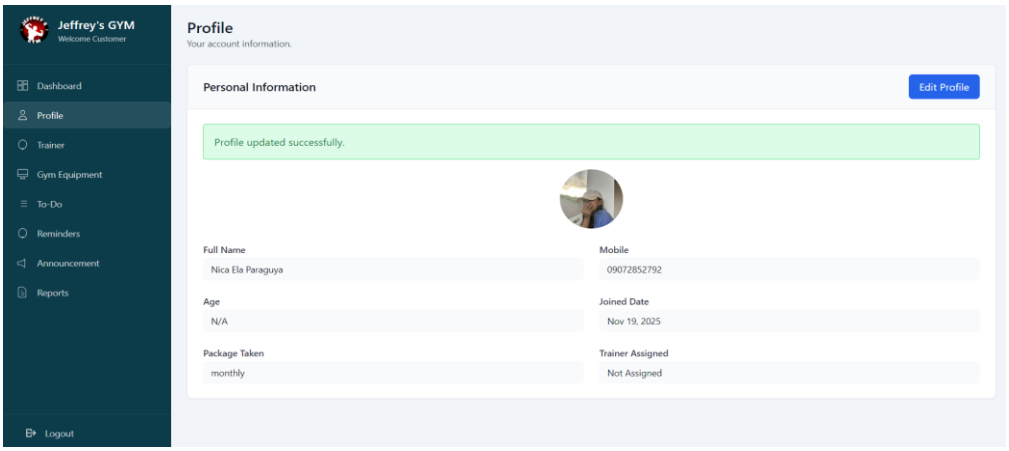


Figure 18. User Interface for Gym Member Application

Figure 18 shows the Profile feature, members can view their personal information such as full name, mobile number, age, membership package, joined date, and assigned trainer. It also allows members to update their profile details using the Edit Profile option, ensuring that their account information remains accurate and up to date for effective communication and membership management.

Evaluation results using TAM and ISO 25010 software product standards to verify its efficacy and quality.

Table 13 reveals consistently positive user perceptions of the gym membership system, with an overall composite mean of 3.55, indicating strong agreement. The results show that users view the system as highly useful in managing their gym activities and supporting their fitness goals. Items related to

system efficiency and ease of accessing gym services such as bookings, payments, and schedules received some of the highest mean scores (3.6), reflecting strong confidence in the system's reliability and convenience. Meanwhile, the statement regarding the system's ability to provide valuable information for fitness goals received a slightly lower mean of 3.3, though it still falls under "Strongly Agree," suggesting minor variation in user opinion. Overall, the composite mean of 3.55 demonstrates that users generally perceive the gym

Table 13.Results of Perceived Usefulness

Perceived Usefulness	Mean	SD	Interpretation
1. The gym membership helps me manage my membership more efficiently.	3.6	±0.48	Strongly Agree
2. Using the system improves my overall gym experience.	3.5	±0.58	Strongly Agree
3. The system make it easier to access gym services (e.g booking , payments and schedule)	3.6	±0.53	Strongly Agree
4. The system provides valuable information that support my fitness goals.	3.3	±0.49	Strongly Agree
5. Overall , I find the gym membership to be useful,	3.5	±0.50	Strongly Agree
COMPOSITE MEAN	**3.55**	**±0.04**	**Strongly Agree**

n=50, Legend: highest to lowest as follows: four (4) – Strongly Agree (3.26 – 4.00), three (3) – Agree (2.51 – 3.25), two (2) – Disagree (1.76 – 2.50), and one (1) – Strongly Disagree (1.0 – 1.75)

Table 14 shows that users strongly agree that the gym membership system is easy to use, as reflected in the composite mean of 3.61. The highest-rated items, with mean scores of 3.6, indicate that users find the system simple to navigate, can accomplish tasks quickly, and are able to use it independently without needing assistance. Convenience also plays a significant role, with respondents agreeing that accessing the system through mobile phones enhances usability mean = 3.5. Likewise, the ease of locating features and information within the platform also

received a strong rating mean = 3.5, showing that the system layout and interface are intuitive. Overall, the results suggest that users perceive the system as user-friendly, accessible, and efficient, contributing to a positive experience in managing their gym activities.

Table 14. Result of Perceived Ease of Use

Perceived Ease of Use	Mean	SD	Interpretation
1. The system is easy to use	3.6	±0.47	Strongly Agree
2. It is more convenient in my end that the gym membership management is accessible through mobile phones.	3.5	±0.50	Strongly Agree
3. I can use the The system without needing from others.	3.6	±0.48	Strongly Agree
4. It is easy to find what I need in the system.	3.5	±0.48	Strongly Agree
5. I can quickly accomplish tasks using the system.	3.6	±0.49	Strongly Agree
COMPOSITE MEAN	**3.61**	**±0.02**	**Strongly Agree**

n=50, Legend: highest to lowest as follows: four (4) – Strongly Agree (3.26 – 4.00), three (3) – Agree (2.51 – 3.25), two (2) – Disagree (1.76 – 2.50), and one (1) – Strongly Disagree (1.0 – 1.75)

Table 15 reveals that users hold a consistently positive attitude toward using the gym membership system, as shown by the composite mean of 3.55, which corresponds to “Strongly Agree.” All items received an identical mean score of 3.5, indicating uniform agreement across different aspects of user attitude. Respondents believe that using the system is a good idea, enjoy managing their gym activities through the platform, and feel more in control of their membership when using it. They also perceive the system as beneficial and supportive of their needs. Overall, the results show that users have a strongly favorable attitude toward the system, demonstrating acceptance, satisfaction, and confidence in its value for gym membership management.

Table 15. Results of Attitude Towards Using the System

Attitude (Towards Using the System)	Mean	SD	Interpretation
1. I have a positive attitude toward using the gym membership system.	3.5	±0.50	Strongly Agree
2. Using of the system is a good idea.	3.5	±0.50	Strongly Agree
3. I enjoy using the system to manage my gym activities.	3.5	±0.50	Strongly Agree
4. I Using the system makes me feel more in control of my membership.	3.5	±0.50	Strongly Agree
5. I believe using the system is more beneficial like me.	3.5	±0.50	Strongly Agree
COMPOSITE MEAN	**3.55**	**±0.00**	**Strongly Agree**

n=50, Legend: highest to lowest as follows: four (4) – Strongly Agree (3.26 – 4.00), three (3) – Agree (2.51 – 3.25), two (2) – Disagree (1.76 – 2.50), and one (1) – Strongly Disagree (1.0 – 1.75)

Table 16 shows that users have a strong intention to continue using the gym membership system, as reflected in the composite mean of 3.60, which corresponds to “Strongly Agree.” The highest-rated items (mean = 3.6) indicate that respondents plan to use the system regularly, intend to rely on it for future membership activities, and are willing to learn more to maximize its features. Users also expressed their willingness to recommend the system to others and encourage fellow gym members to adopt it, both with mean scores of 3.5. These results demonstrate a high level of user acceptance and commitment, highlighting the system’s positive impact on user behavior and their intentions to integrate it into their ongoing gym routines.

Table 16. Results of Behavior

Behavior (Intention to Use the System)	Mean	SD	Interpretation
1. I intend to use the gym membership system regularly.	3.6	±0.49	Strongly Agree
2. I will recommend the system to other gym member.	3.5	±0.50	Strongly Agree
3. I plant to use the system for managing my membership activities in the future.	3.6	±0.49	Strongly Agree
4. I am willing to learn more about the system to use it effectively.	3.6	±0.51	Strongly Agree
5. I will encourage other member to use the system.	3.5	±0.50	Strongly Agree
COMPOSITE MEAN	**3.60**	**±0.09**	**Strongly Agree**

n=50, Legend: highest to lowest as follows: four (4) – Strongly Agree (3.26 – 4.00), three (3) – Agree (2.51 – 3.25), two (2) – Disagree (1.76 – 2.50), and one (1) – Strongly Disagree (1.0 – 1.75)

The table 17 show results presented in the table show that all Technology Acceptance Model (TAM) constructs obtained high mean scores, each interpreted as Strongly Agree. Perceived Ease of Use recorded the highest mean (M = 3.61, SD = ±0.02), indicating that respondents strongly agreed that the system is easy to learn and operate. Behavioral Intention to Use followed closely with a mean of 3.60 (SD = ±0.09), reflecting a strong intention among users to continue using the system. Meanwhile, Perceived Usefulness and Attitude Toward Using both obtained a mean of 3.55, suggesting that respondents strongly perceived the system as beneficial and maintained a positive attitude toward its use.

The composite mean of 3.60 (SD = ±0.09) further reinforces the overall Strongly Agree interpretation, demonstrating a high level of acceptance of the system among the 50 respondents. Based on the given scale, all mean values fall within the highest range (3.26–4.00), indicating that users not only find the system useful and easy to use but also exhibit favorable attitudes and strong intentions to

adopt and continue using it. These findings suggest that the system is highly acceptable and well-received by its intended users.

Table 17 Summary of TAM Evaluation Results

TAM Construct	Mean	SD	Interpretation
Perceived Usefulness	3.55	±0.04	Strongly Agree
Perceived Ease of Use	3.61	±0.02	Strongly Agree
Attitude Toward Using	3.55	±0.00	Strongly Agree
Behavioral Intension to Use	3.60	±0.09	Strongly Agree
COMPOSITE MEAN	**3.60**	**±0.09**	**Strongly Agree**

n=50, Legend: highest to lowest as follows: four (4) – Strongly Agree (3.26 – 4.00), three (3) – Agree (2.51 – 3.25), two (2) – Disagree (1.76 – 2.50), and one (1) – Strongly Disagree (1.0 – 1.75)

Project Testing and Debugging

Unit Testing

The table 18 show unit testing conducted for the Digital Workout: Progressive Web-Application for Gym Membership and Monitoring ensured that all major modules functioned effectively and met their intended purposes. Each component was carefully tested to guarantee reliability, usability, and system integrity for both gym members and administrators. Login Module was tested for security and functionality. Members and administrators were able to access their respective dashboards using valid credentials, while unauthorized logins were blocked. This ensured the protection of user information and maintained system security.

Account Module allowed members and administrators to manage their profiles, update personal information, and set preferences. Testing confirmed that all profile updates were saved accurately and that account settings were fully functional without errors. equipment Module was tested to ensure proper tracking of gym equipment. Administrators could add, update, and monitor equipment status, while members could view available equipment. Tests verified that equipment data was accurately recorded and updated in real-time. Attendance Module enabled accurate manual check-ins and check-outs for members. Members' attendance history was correctly recorded, and session attendance tracking worked seamlessly. Notifications for attendance updates were delivered promptly. payment Module supported secure transactions via GCash and cash payments. Testing confirmed that payments were processed correctly, membership statuses were updated immediately, and administrators could access accurate transaction records.

Report Module generated comprehensive and visually engaging analytics on member activity, attendance trends, equipment usage, and payments. Testing verified that all reports were accurate, updated in real-time, and accessible to administrators for decision-making. the unit testing results confirmed that the system functions reliably, maintains data security, and provides a user-friendly experience.

Table 18. Unit Testing for Digital Workout System

Tested Case	Expected Results	Actual Results	Response Time	Remarks
Test login functionality for members and administrators using valid and invalid credentials	Only registered users can log in; unauthorized access blocked	Member and admin logins worked correctly; unauthorized users denied access	0.45 s	Passed
Verify member registration and profile creation	Members can create an account and access dashboard	Registration and dashboard access successful	1.10 s	Passed
Update member profile and account settings	Changes saved accurately	Profile updates reflected correctly	0.60 s	Passed
Add, update, and view gym equipment status	Equipment data recorded and updated correctly	Equipment information updated and displayed accurately	0.75 s	Passed
Record member check-ins and check-outs	Attendance logged accurately	Check-ins and check-outs recorded correctly	0.25 s	Passed
Process GCash and cash payments for memberships	Payments recorded; membership status updated	Transactions processed successfully; status updated	1.25 s	Passed
Generate analytics and visual reports on members, attendance, and payments	Reports accurately display trends and activity	Reports generated correctly with all relevant data	1.00 s	Passed
Test data protection for login, account, and payment modules	User credentials and payment/member data protected from unauthorized access	All sensitive data encrypted; no breaches occurred	0.50 s	Passed

Integration Testing

The table 19 show Integration testing was conducted to verify that all modules of the Digital Workout system worked together seamlessly. The purpose was to ensure smooth communication between modules such as Membership, Trainer, Attendance, Payment, and Data Analytics, guaranteeing that the entire system functioned cohesively. During testing, interactions between modules were examined thoroughly. For example, when a member registered and selected a plan in the Membership Module, the Attendance Module accurately tracked check-ins according to the selected plan. Payments made via GCash or cash immediately reflected in the Payment Module and updated membership status in the Member Dashboard. Notifications were correctly triggered to inform members of successful registration, upcoming sessions, or payment confirmations.

The integration between the Trainer Module and Attendance Module allowed administrators to schedule training sessions, monitor trainer availability, and track member participation in real-time. Additionally, the Data Analytics Module successfully pulled information from all other modules to generate comprehensive reports on member activity, attendance trends, trainer performance, and financial data. Integration testing confirmed that the system's modules interacted efficiently without errors, ensuring a reliable and unified user experience for both members and administrators. All interfaces were smooth, and no data inconsistencies were observed.

Table 19. Integration Testing for Digital Workout System

Tested Case	Expected Results	Actual Results	Response Time	Remarks
Account Module	Users can securely log in and manage account information	Account access and updates functioned correctly without errors	0.45 s	Passed
Manage Member Module	Members can be added, updated, and viewed accurately	Member records were saved and displayed correctly	0.60 s	Passed
Equipment Module	Gym equipment details can be added, updated, and monitored	Equipment information updated and shown accurately	0.55 s	Passed
Trainers Module	Trainer profiles and schedules are properly managed	Trainer data and schedules reflected correctly	0.70 s	Passed
Attendance Module	Member check-ins and check-outs are recorded accurately	Attendance records were logged correctly in real time	0.80 s	Passed
Payment Module	Payments are processed and membership status updates automatically	Payment transactions completed successfully and status updated	1.10 s	Passed
Announcement Module	Announcements can be created and viewed by members	Announcements displayed correctly to all users	0.35 s	Passed
Report Module	System generates accurate reports for members, attendance, and payments	Reports generated correctly with complete and accurate data	1.25 s	Passed

System Testing

The table 20 show browser compatibility testing results indicate that the system performs efficiently and consistently across Google Chrome, Mozilla Firefox, and Microsoft Edge. Core functionalities such as User Login and Authentication, Account Management, and Member Management showed minimal variation in response times among the three browsers, with all processes completing in approximately one to one and a half seconds. This consistency suggests that the system is stable and responsive regardless of the browser used, ensuring a reliable user experience for both administrators and members.

Additionally, transaction-related and operational features demonstrated satisfactory performance across all browsers. Payment Processing (Cash and GCash) recorded slightly higher response times compared to other modules, particularly in Firefox; however, all results remained within acceptable performance limits. Modules such as Attendance Recording, Equipment Management, and Announcement Management exhibited faster response times, generally completing in under one second. Overall, the findings confirm that the system is compatible with major web browsers and capable of delivering timely responses across its key functionalities, supporting seamless operation in different web environments.

Table 20 Browser Compatibility Testing Result

Tested Case	Chrome Response Time	Firefox Response Time	Microsoft Edge Response Time
User Login and Authentication	1.25 s	1.30 s	1.28 s
Account Management (Admin Access)	1.26 s	1.37 s	1.25 s
Member Management	1.25 s	1.38 s	1.26 s
Payment Processing (Cash & G-cash)	1.50 s	1.65 s	1.22 s
Attendance Recording	0.90 s	0.72 s	0.88 s
Equipment Management	0.95 s	0.68 s	0.66 s
Announcement Management	0.60 s	0.70 s	0.68 s
Report Management	0.85 s	1.38 s	0.86

Acceptance Testing

Table 21 presents the results of the acceptance testing of the system based on various quality characteristics. The findings indicate that the system achieved high mean scores across all evaluated criteria, reflecting a high level of user acceptance. Among the characteristics, Flexibility obtained the highest mean score (M = 4.69, SD = 0.04), followed by Maintainability (M = 4.65, SD = 0.22) and

Security (M = 4.54, SD = 0.17), all of which were interpreted as Excellent. These results suggest that users strongly perceived the system as adaptable to changing requirements, easy to maintain, and secure in handling data and system operations.

also received Excellent interpretations, indicating that the system effectively meets user requirements, supports smooth user interaction, operates reliably, and ensures safe usage. Although Performance Efficiency (M = 4.08, SD = 0.26) and Compatibility (M = 4.12, SD = 0.10) were interpreted as Good, their mean scores remain high, suggesting satisfactory performance across different environments and acceptable system responsiveness. The overall mean of 4.40 (SD = 0.08), interpreted as Excellent, confirms that the system is highly acceptable among the 50 respondents, demonstrating its readiness for actual deployment and use.

Table 21. Acceptance Testing of the System

Characteristics	Mean	SD	Interpretation
Functional Suitability	4.27	0.24	Excellent
Performance Efficiency	4.08	0.26	Good
Compatibility	4.12	0.10	Good
Interaction Capability	4.38	0.20	Excellent
Reliability	4.43	0.24	Excellent
Security	4.54	0.17	Excellent
Maintainability	4.65	0.22	Excellent
Flexibility	4.69	0.04	Excellent
Safety	4.45	0.24	Excellent
Overall Mean	**4.40**	**0.08**	**Excellent**

n=50

CHAPTER 5

SUMMARY, CONCLUSION AND RECOMMENDATION

This chapter gives a summary of the capstone project. It explains the conclusions from the study and provides recommendations that helped make the project successful.

Summary

The study developed Digital Workout: A Progressive Web Application for Gym Membership and Monitoring to provide a modern and efficient solution for managing gym operations. Traditional gym processes—such as manual encoding of member records, attendance logging, payment tracking, and monitoring of workout progress—often result in delays, inaccuracies, and inconvenience for both gym staff and members. To address these issues, the researchers designed a PWA-based system that can be accessed on any device, offering flexibility, faster transactions, and real-time monitoring.

The system includes key features such as member registration, attendance tracking using digital logs, membership payment monitoring, and a dashboard for viewing member information and progress. Gym staff can easily update records, while members can view their personal data and monitor their activity anytime. The development and evaluation process followed industry standards to ensure that the system is user-friendly, functional, secure, and reliable. Based on feedback from users and evaluators, the Digital Workout PWA successfully demonstrates its ability to improve gym operations and support member engagement.

Conclusion

In conclusion, the responsive web-based gym membership and monitoring system successfully achieved the following objectives:

To create an accessible and responsive web-based platform that provides a seamless user experience across all devices (desktop, tablet, and mobile) without the need for native app installation.

The project successfully delivered a Responsive web-based that is fully responsive and accessible across desktop, tablet, and mobile devices. The system was built using modern web technologies such as HTML, CSS, JavaScript, and Laravel 9, ensuring a consistent and intuitive user interface. Testing across multiple browsers (Chrome, Firefox, Edge) and devices confirmed that the platform operates smoothly without requiring installation, providing gym members and administrators with flexible, on-the-go access to gym management features.

To implement core functional modules for user accounts, membership management, payment processing and progress monitoring.

The Gym Management System successfully meets the objective of automating workout progress monitoring, membership tracking, and report generation. The system enables administrators and trainers to monitor members' workout progress, attendance, eliminating the need for manual tracking. Automated reports provide clear summaries of member activities, progress trends, and membership status, which help gym management make informed decisions regarding training programs and services. These features enhance operational

efficiency and ensure that members receive timely and accurate feedback on their fitness progress, contributing to improved engagement and satisfaction.

To evaluate the acceptability of the developed system using ISO 25010 and Technology Acceptance Model (TAM)

The evaluation of the Gym Management System using ISO 25010 and the Technology Acceptance Model (TAM) confirmed its efficiency, effectiveness, and usability. Based on the ISO 25010 standards, the system demonstrated strong performance in terms of functional suitability, usability, reliability, and security, indicating that it meets the quality requirements expected of a modern web-based application. The TAM results further revealed a high level of user acceptance, showing that users found the system easy to use, useful in managing gym-related tasks, and beneficial to overall gym operations. These positive evaluation results validate the system's readiness for real-world implementation and its potential to improve the management and monitoring processes within the gym.

Recommendation

1. Strengthen Security Measures Implement more advanced authentication methods, such as two-factor authentication, data encryption, and regular security audits to ensure maximum protection of user information.

2. Add Comprehensive Workout Tracking Tools Include features such as personalized fitness plans, exercise tutorials, automated progress charts, and trainer feedback modules to further support members in achieving their fitness goals.

3. Provide Training Sessions for Staff and Members Offer basic system training or user guides to help ym staff and members maximize the features of the PWA and ensure smooth adaptation.

4. Conduct Regular Maintenance and Updates Perform scheduled upgrades to fix bugs, improve system performance, and introduce new features based on user feedback.

REFERENCES

Bakar N. & Said M. (2019). The Development of an Online Gym Booking System (OGBS) using Online Membership Business Model https://www.warse.org/IJATCSE/static/pdf/file/ijatcse11815sl2019.pdf

Chokhandre S. et. Al. (2025). Gym Management System https://www.ijsrtjournal.com/article/Gym+Management+System

Ebrahimi A. et Al. (2017). Gym-Suite an Extensible Platform for Gyms Management.

Delgado R. H. (2021). Gym-Suite An extensible platform for gyms management. https://www.proquest.com/openview/2ff64c6aaafee3042297b23410a6ba84/1?cbl=2026366&diss=y&pq-origsite=gscholar.

Gamage T. et. Al. (2022). Gym Management in Sri Lanka https://www.indianjournals.com/ijor.aspx?target=ijor:ijemr&volume=12&issue=5&article=009

Hadawale S. et. Al. (2025). The Role of Web-Based Gym Management Systems in Enhancing User Experience https://ijrpr.com/uploads/V6ISSUE2/IJRPR38945.pdf

Hambali M. H. et. Al. (2024). Design and Evaluation of a Web-based Membership Management System https://e-journal.uum.edu.my/index.php/jdsd/article/view/23015

Hui L. & Teo P. (2024). An Implementation of Digital Platform to Enhance the Appointment Scheduling System

https://hrmars.com/papers_submitted/20635/an-implementation-of-digital-platform-to-enhance-the-appointment-scheduling-system.pdf

Jain A. (2015). A smart gym framework: Theoretical approach https://ieeexplore.ieee.org/abstract/document/7434423

Karmakar A. (2022). Gym Management System https://www.studocu.com/row/document/hope-university-college/computer-science/aju211506-cs-vp1/108681287

Khattri V. et. Al. (2024). Online Payment Transaction System https://link.springer.com/chapter/10.1007/978-981-97-4888-4_2

Kendiga J. (2017). Taurama Fitness Centre System https://repository.unikom.ac.id/53919/

Kotzé M. A. (2017). A framework for the steps towards a smart gym https://www.researchgate.net/publication/335453284_A_framework_for_the_steps_towards_a_smart_gym

Kumar Dinesh A.V. et. Al. (2020). Smart Gym Management System https://ijsret.com/wp-content/uploads/2020/05/IJSRET_V6_issue3_493.pdf

Lakshitha K. & Sirisuriya S. (2021). An Automated Platform to Manage Customer Relationship in a Gymnasium http://ir.kdu.ac.lk/handle/345/5242

Löthman & Liljeblad. (2021). Committed to Exercise: A qualitative study on the persuasive effect and potential of a gym's online booking system https://www.diva-portal.org/smash/record.jsf?pid=diva2:1591125

Monir A. & Jannatun N. (2016). A Project Paper on Smart Gym Management System http://103.133.167.11:8080/bitstream/handle/123456789/2075/Monir_Ahmed.pdf?sequence=1&isAllowed=y

Nivedita V. et. Al (2025). Web-based fitness trainer and client's management system https://www.researchgate.net/publication/390409310_Web-based_fitness_trainer_and_client's_management_system

Orange L. M. (2017). Gym Membership Wellness Program as a Cost Benefit to Health Insurance Companies https://www.proquest.com/openview/c2674ef3b18f9a6c048ea8f0411f109d/1?cbl=18750&pq-origsite=gscholar

Sembiring R. (2023). Design Of A Web-Based Membership Data Processing System At Vizta Gym Using A Prototype Method

Shakoor M. et. Al. (2018). Database and Transactions Management System for a Smart Gym: LAYYAH FITNESS CENTER 340491607_DATABASE_AND_TRANSACTIONS_MANAGEMENT_SYSTEM_FOR_A_SMART_GYM_LAYYAH_FITNESS_CENTER

Shan S. R. (2023). Gym ERP Management System using Machine Learning https://openurl.ebsco.com/EPDB%3Agcd%3A15%3A17097775/detailv2?sid=ebsco%3Aplink%3Ascholar&id=ebsco%3Agcd%3A174054467&crl=c&link_origin=scholar.google.com

Siagian H. (2023). Web-Based Information System for Managing Gymnasium Membership http://repository.president.ac.id/handle/123456789/12060

Suryawanshi A. et al. (2024). Gym Management System https://www.ijcstjournal.org/volume-12/issue-4/IJCST-V12I4P4.pdf

Tsai H. et Al . (2025). A New Scheme Modeling Gym Membership Transaction with NFT Systems https://link.springer.com/chapter/10.1007/978-981-97-7786-0_13

Vaishnavi Reddy G. et. Al. (2024). Digital Gym Management Systems: User Experience & Efficiency Study https://ijrpr.com/uploads/V5ISSUE3/IJRPR23946.pdf

APPENDICES

APPENDIX A

Letter of Approval

Republic of the Philippines
Laguna State Polytechnic University
Province of Laguna

CERTIFICATE OF APPROVAL

This is to certify that after having satisfactorily completed all the requirements of the capstone project entitled:

DIGITAL WORKOUT: DEVELOPMENT OF PROGRESSIVE WEB - APPLICATION FOR GYM MEMBERSHIP AND MONITORING

NICA ELA D. PARAGUYA, JHON RIO GONZALES AND RODELIO V. RAQUEL have been granted permission to proceed with his final oral presentation.

As their Research Adviser, I confirm that the proponents had met the academic standards and requirements set forth by the department, and therefore, I approve their request to conduct the final oral presentation.

JONALY JOY LABAYNE
Capstone Project Adviser

APPROVAL SHEET

The Capstone Project entitled "**DIGITAL WORKOUT: DEVELOPMENT OF RESPONSIVE WEB-BASED FOR GYM MEMBERSHIP AND MONITORING"** prepared and submitted by **Nica Ela D. Paraguya, Rodelio Raquel** and **Jhon Rio Gonzales** in partial fulfillment of the requirements for the degree Bachelor of Science in Information Technology specialized in Service Management Program, is hereby recommended for approval and acceptance.

JONALYN JOY LABAYNE
Adviser

Approved and Accepted by the Committee on Oral Examination with a grade of _____.

MIKE PHILIP T. RAMOS
Member

SHERWIN B. SAPIN, EdD
Member

JOCELYN O. PADALLAN
Member

JEFFERSON L. LERIOS, DIT
Member

Accepted in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Technology specialized in Service Management Program.

LOYD S. ECHALAR
Associate Dean, College of Computer Studies

OFELIA B. MANINGAS, EdD.
Chairperson, Research and Development Office

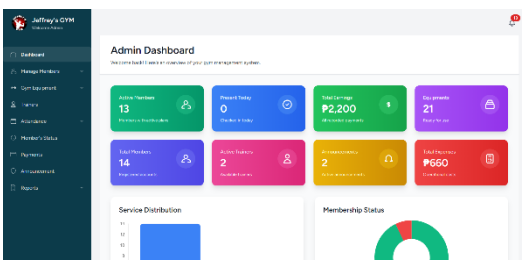
_____
Date Signed

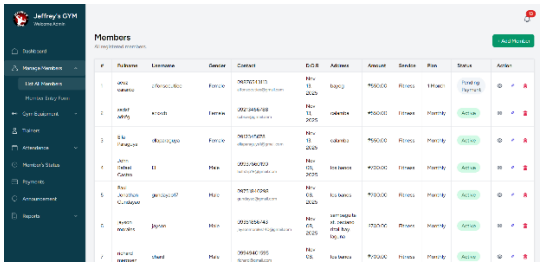
RESEARCH CONTRIBUTION NO.	

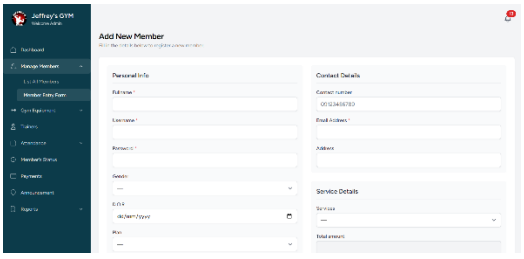
APPENDIX B

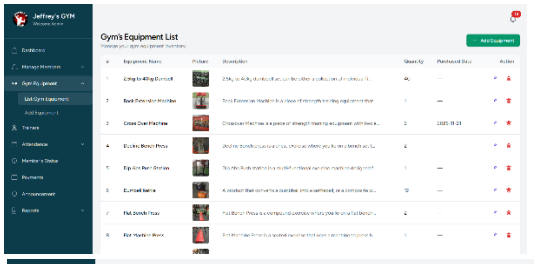
System Interface Screenshots

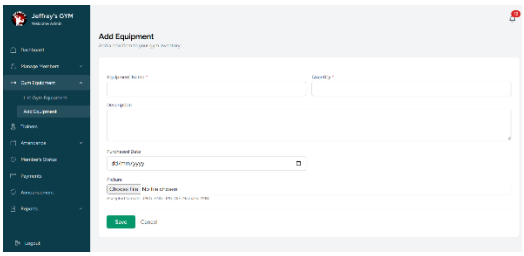
Administrator Interface

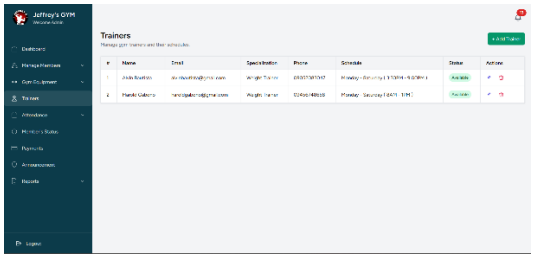


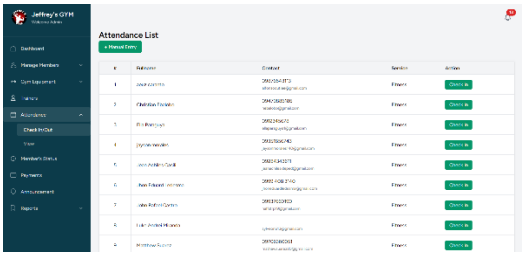


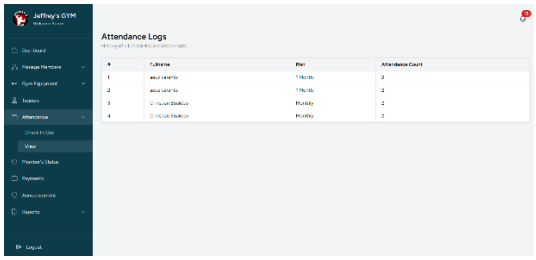


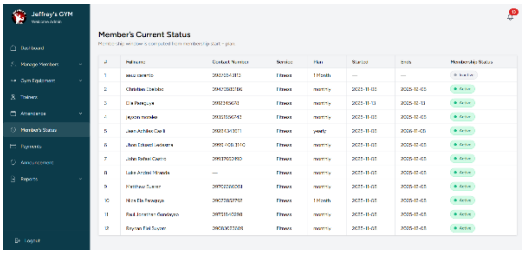


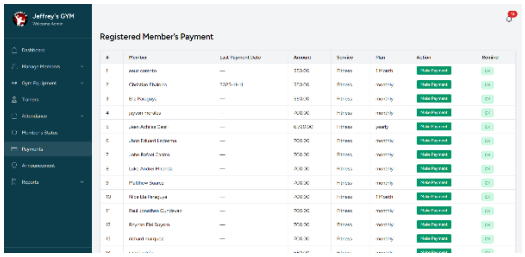


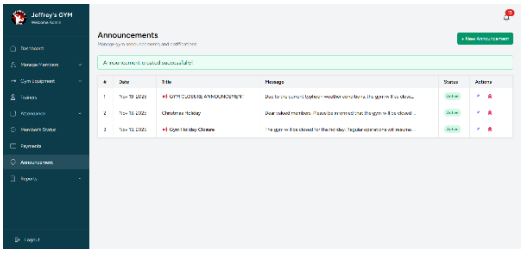


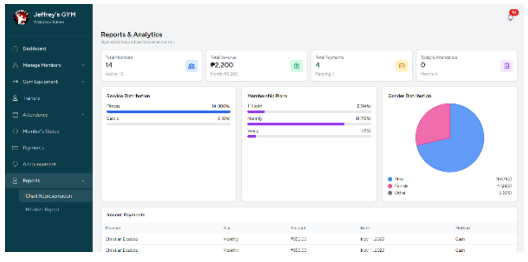




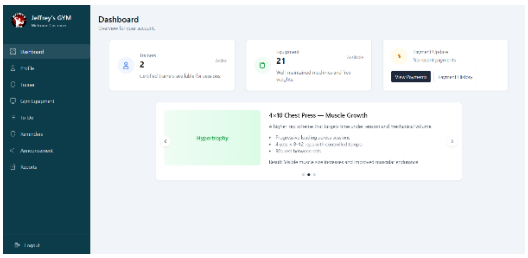


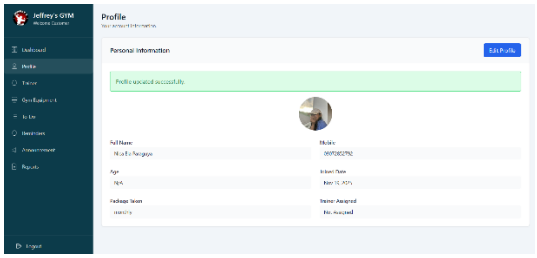


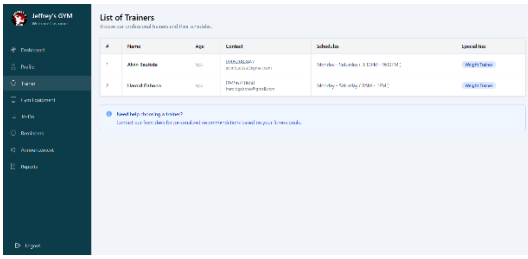


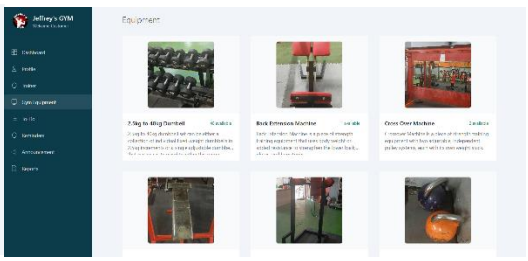


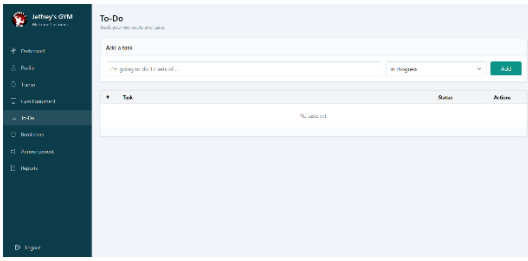
User Interface

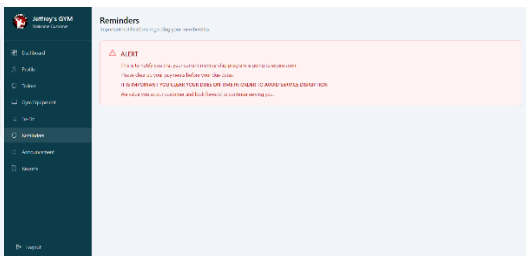


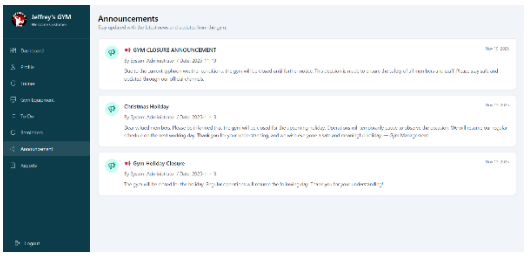


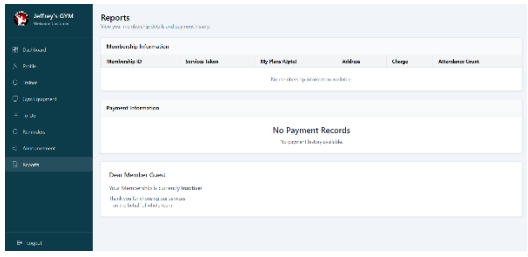








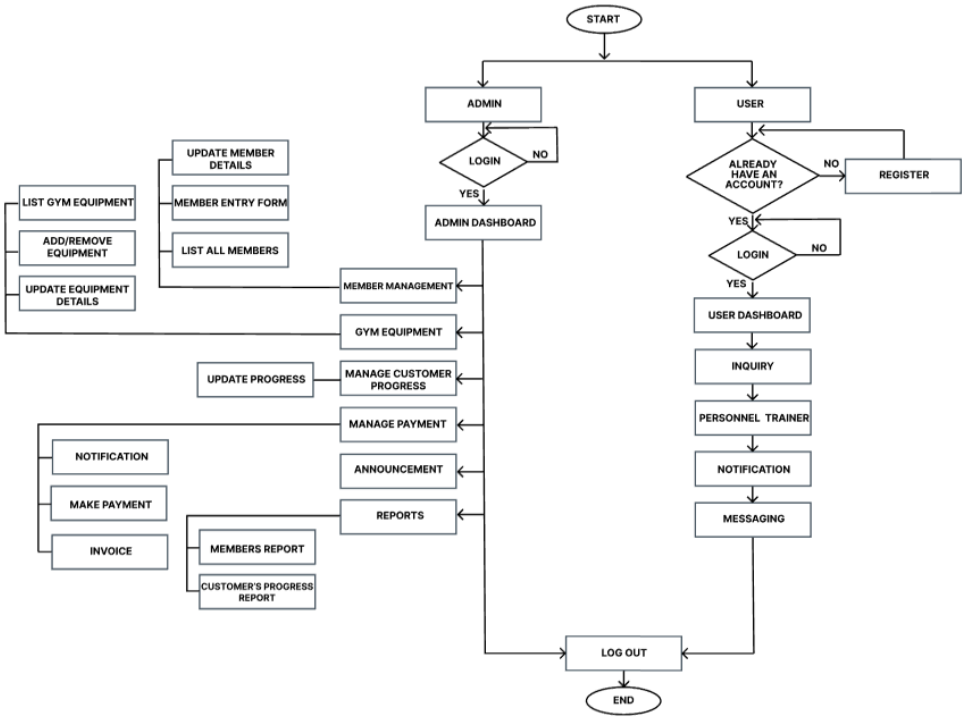




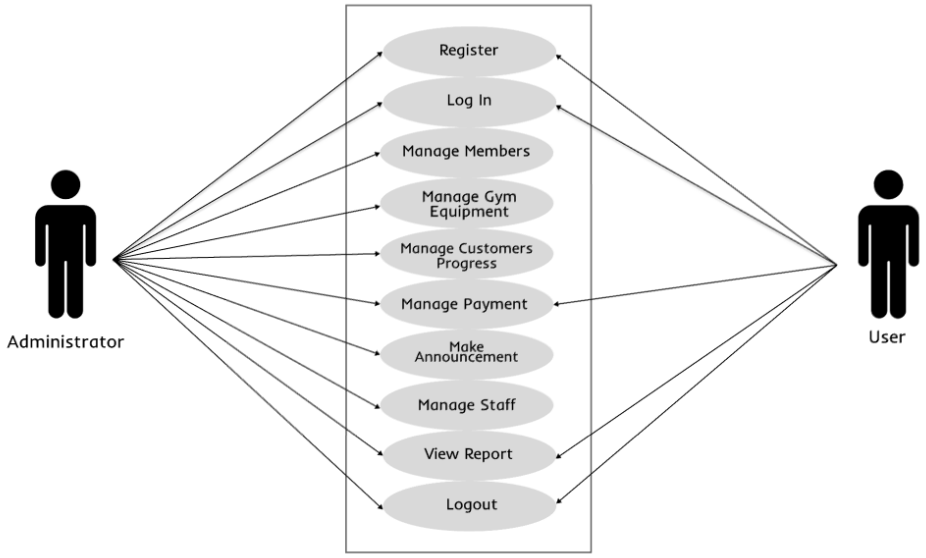
APPENDIX C

Diagram

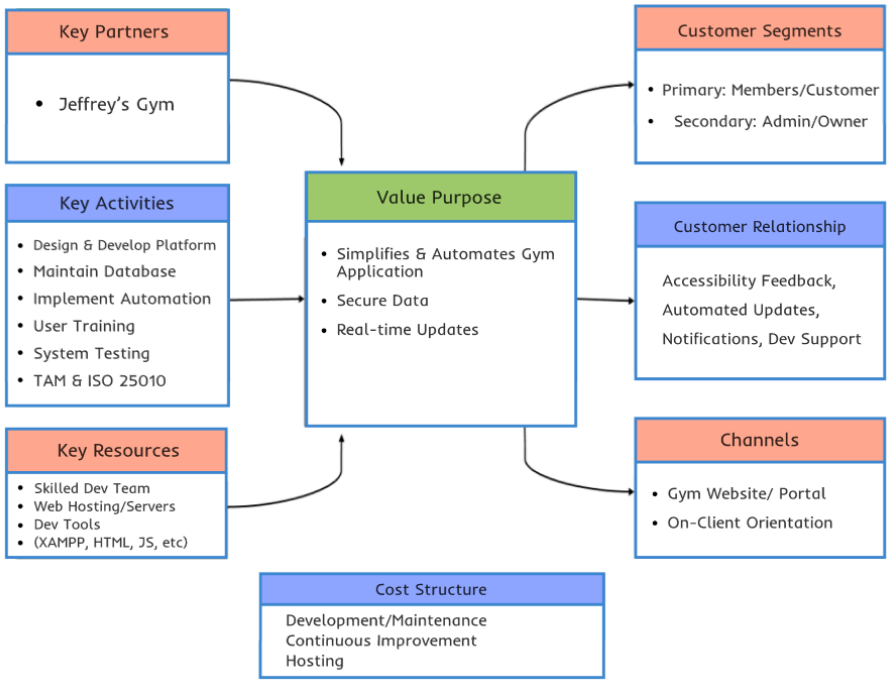
Flow Chart



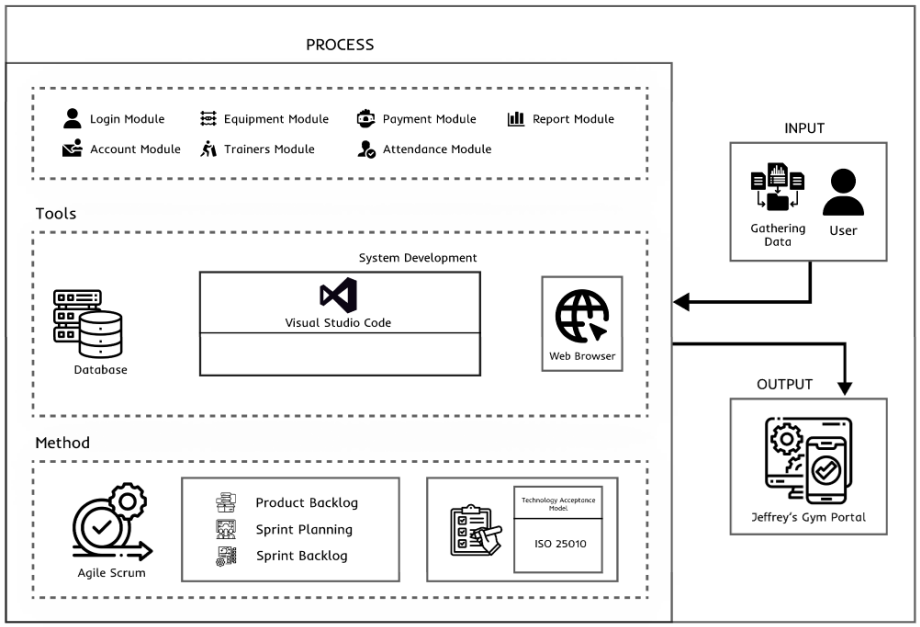
Use Case Diagram



Business Model



Conceptual Framework



APPENDIX D

Survey Questionnaire / Evaluation Tool

DIGITAL WORKOUT: DEVELOPMENT OF RESPONSIVE WEB-APPLICATION FOR GYM MEMBERSHIP MANAGEMENT AND MONITORING

Dear respondent,

We are fourth-year students currently pursuing the Bachelor of Science in Information Technology, specializing in Service Management Program. As part of our academic requirements, we have developed an **"Digital workout: Development of Responsive Web-Application for Gym Membership Management and Monitoring"**, which adheres to the ISO/IEC 25010 Software Quality Model**.**

To evaluate the system's effectiveness and overall quality, we are conducting an assessment focusing on key aspects such as usability, functionality, security, and user experience **.**Your participation in this evaluation is highly appreciated. Kindly assess each item based on your experience with the system, using the 5- point rating scale provided below. Your responses will be treated with the utmost confidentiality and will be used solely for academic purpose.Thank you for your time and cooperation.

RESEARCHER: Nica Paraguya, Rodelio Raquel & Jhon Rio Gonzales.

Direction: Check the rubric that corresponds to your evaluation

Functional Suitability

Subcategory	Poor (1)	Need Improvements (2)	Fair (3)	Good (4)	Excellent (5)
Functional Completeness	Lacks main gym features.	Some key features missing.	Basic features work; some incomplete.	Most features work well.	All features complete and integrated.
Functional Correctness	Many errors and unreliable results.	Some inaccurate outputs.	Occasional minor errors.	Mostly accurate and stable.	Fully accurate and error-free.
Functional Appropriateness	Functions not useful for users.	Limited usability and efficiency.	Moderately useful; needs improvement.	Useful and supports gym tasks.	Highly relevant, user-friendly, and efficient.

Reliability

Subcategory	Poor (1)	Need Improvements (2)	Fair (3)	Good (4)	Excellent (5)
Recoverability	System fails to recover from crashes or data loss.	Limited recovery options; manual fixes needed.	Partial recovery possible but not consistent.	System recovers quickly from most errors.	Full recovery from failures with no data loss.
Maturity	System frequently fails; unstable.	Occasional failures; needs stabilization.	Generally stable but with some issues.	Stable performance with rare minor issues.	Fully stable and reliable under all conditions.
Availability	.System often down or inaccessible.	Frequent downtime; uptime needs improvement.	Acceptable uptime but occasional outages.	High uptime with minimal interruptions.	Always available and responsive.
Fault Tolerance	Crashes easily when errors occur.	Limited ability to handle errors.	Can handle minor faults with some impact.	Handles faults smoothly with little disruption.	Continues running effectively even under faults or stress

Performance Efficiency

Subcategory	Poor (1)	Need Improvements (2)	Fair (3)	Good (4)	Excellent (5)
Time Behavior	System is slow; long response times.	Noticeable delays under normal use.	Acceptable response times; occasional lag.	Fast response under most conditions.	Quick and consistent response under all conditions.

Resource Utilization	Consumes excessive memory/CPU; inefficient.	High resource use; some inefficiency.	Moderate resource usage; acceptable performance.	Efficient use of resources; minimal wastage.	Optimized use of memory/CPU; highly efficient.
Capacity	.Cannot handle expected number of users or data.	Limited user/data capacity; performance drops under load.	Handles normal load; may struggle with peak usage.	Supports expected load comfortably.	Easily supports high number of users and large data volumes.

Usability

Subcategory	Poor (1)	Need Improvements (2)	Fair (3)	Good (4)	Excellent (5)
Appropriateness Recognizability	Functions not aligned with user needs.	Partially supports user tasks.	Moderately suitable for tasks.	Mostly supports user tasks.	Fully appropriate for user goals.
User Engagement	System is boring; users lose interest quickly.	Limited engagement; features not motivating.	Moderately engaging; some features keep users interested.	System is engaging; users interact regularly.	Highly engaging; motivates consistent and meaningful interaction.
Learnability	.Hard to learn; requires much effort.	Learning requires guidance.	Users learn with some effort.	Easy to learn; minor guidance needed.	Very easy to learn; intuitive design.
Operability	Difficult to operate; inefficient workflow.	Operation requires workarounds.	Usable but not smooth.	Smooth operation; minimal effort.	Very efficient and intuitive operation.
User Error Protection	System prone to frequent user errors; no safeguards.	Some safeguards, but errors possible.	Occasional errors; some protection present.	Good error prevention; minor issues.	Strong error prevention and recovery mechanisms.
Inclusivity	Not usable by diverse users; excludes many groups.	Some accessibility and inclusivity features; gaps remain.	Moderately inclusive; works for most users with effort.	Inclusive for most users; minimal barriers.	Fully inclusive; accessible and usable by all intended users.

TECHNOLOGY ACCEPTANCE MODEL QUESTIONNAIRE

Dear respondent,

We are students of the Laguna State Polytechnic University – Los Baños Campus under the College of Computer Studies conducting development on the project entitled **"Digital workout: Development of Responsive Web-Application for Gym Membership Management and Monitoring"** We would like to seek your opinion on the acceptability of the said project by scoring the statements below by checking the appropriate numbers, wherein 4 = Strongly Agree; 3 = Agree; 2 = Disagree; and 1 = Strongly Disagree. Rest assured that any information that you provide will be treated with utmost confidentiality (RA 10173. Data Privacy Act) and will be used for research purposes only. Thank you for your cooperation.

Nica Ela D. Paraguya
Jhon Rio Gonzales
Rodelio V. Raquel

NAME OF RESPONDENT: _____

Perceived Usefulness	**4**	**3**	**2**	**1**
1. The gym membership system helps me manage my membership more efficiently. *Nakakatulong ang gym membership system upang mas madaling pamahalaan ang aking membership*.				
2.Using the system improves my overall gym experience.. *Pinapahusay ng paggamit ng systema ang aking karanasan sa gym.*				
3. The system makes it easier to access gym services (e.g., booking, payments, schedules). *Pinapadali ng systema ang pag-access sa mga serbisyo ng gym (hal. pag-book, bayad, iskedyul).*				
4. The system provides valuable information that supports my fitness goals. *Nagbibigay ang systema ng mahalagang impormasyon na tumutulong sa aking fitness goals.*				

5. Overall, I find the gym membership system to be useful. *Sa kabuuan, kapaki-pakinabang para sa akin ang gym membership system.*				
Perceived Ease of Use				
1. The system is easy to use. *Madaling gamitin ang system.*				
2. Learning to operate the system is easy for me. *Madali para sa akin ang pag-aralan kung paano gamitin ang systema.*				
3. I can use the system without needing help from others. *Kaya kong gamitin ang systema nang hindi nangangailangan ng tulong mula sa iba..*				
4. It is easy to find what I need in the system. *Madali kong mahanap ang mga kailangan ko sa loob ng systema*.				
5. I can quickly accomplish tasks using the system. *Mabilis kong natatapos ang mga gawain gamit ang systema.*				
Attitude (Towards Using the System)				
1. I have a positive attitude toward using the gym membership system. *Maganda ang pananaw ko sa paggamit ng gym membership system.*				
2. Using the system is a good idea. *Magandang ideya ang paggamit ng systema.*				
3. I enjoy using the system to manage my gym activities. *Nasisiyahan ako sa paggamit ng systema para pamahalaan ang aking mga aktibidad sa gym*.				
4. Using the system makes me feel more in control of my membership. *Mas nararamdaman kong kontrolado ko ang aking membership sa pamamagitan ng systema.*				

5. I believe using the system is beneficial for gym members like me. *Naniniwala akong kapaki-pakinabang ang systema para sa mga miyembro ng gym tulad ko.*				
Behavior (Intention to Use the System)				
1. I intend to use the gym membership system regularly *Plano kong gamitin nang regular ang gym membership system.*				
2. I will recommend the system to other gym members. *Ire-rekomenda ko ang systema sa iba pang miyembro ng gym.*				
3. I plan to use the system for managing my membership				
activities in the future. *Plano kong gamitin ang systema para pamahalaan ang aking membership sa hinaharap.*				
4. I am willing to learn more about the system to use it effectively. *Handa akong matuto pa tungkol sa systema para magamit ito nang maayos.*				
5. II will encourage other members to use the system. *Hihikayatin ko ang ibang miyembro na gamitin ang systema.*				

_____

Signature of the Respondent

APPENDIX E

Relevant Sources

```php
<?php

namespace App\Console;

use Illuminate\Console\Scheduling\Schedule;
use Illuminate\Foundation\Console\Kernel as ConsoleKernel;

class Kernel extends ConsoleKernel
{
    /**
     * Define the application's command schedule.
     *
     * @param  \Illuminate\Console\Scheduling\Schedule  $schedule
     * @return void
     */
    protected function schedule(Schedule $schedule)
    {
        // $schedule->command('inspire')->hourly();
    }

    /**
     * Register the commands for the application.
     *
     * @return void
     */
    protected function commands()
    {
        $this->load(__DIR__.'/Commands');

        require base_path('routes/console.php');
    }
}
```

```php
<?php

use Illuminate\Support\Str;

return [

    'connections' => [

        'sqlite' => [
            'driver' => 'sqlite',
            'url' => env('DATABASE_URL'),
            'database' => env('DB_DATABASE', database_path('database.sqlite')),
            'prefix' => '',
            'foreign_key_constraints' => env('DB_FOREIGN_KEYS', true),
        ],
```

```
        'mysql' => [
            'driver' => 'mysql',
            'url' => env('DATABASE_URL'),
            'host' => env('DB_HOST', '127.0.0.1'),
            'port' => env('DB_PORT', '3306'),
            'database' => env('DB_DATABASE', 'forge'),
            'username' => env('DB_USERNAME', 'forge'),
            'password' => env('DB_PASSWORD', ''),
            'unix_socket' => env('DB_SOCKET', ''),
            'charset' => 'utf8mb4',
            'collation' => 'utf8mb4_unicode_ci',
            'prefix' => '',
            'prefix_indexes' => true,
            'strict' => true,
            'engine' => null,
            'options' => extension_loaded('pdo_mysql') ? array_filter([
                PDO::MYSQL_ATTR_SSL_CA => env('MYSQL_ATTR_SSL_CA'),
            ]) : [],
        ],

        'pgsql' => [
            'driver' => 'pgsql',
            'url' => env('DATABASE_URL'),
            'host' => env('DB_HOST', '127.0.0.1'),
            'port' => env('DB_PORT', '5432'),
            'database' => env('DB_DATABASE', 'forge'),
            'username' => env('DB_USERNAME', 'forge'),
            'password' => env('DB_PASSWORD', ''),
            'charset' => 'utf8',
            'prefix' => '',
            'prefix_indexes' => true,
            'search_path' => 'public',
            'sslmode' => 'prefer',
        ],

        'sqlsrv' => [
            'driver' => 'sqlsrv',
            'url' => env('DATABASE_URL'),
            'host' => env('DB_HOST', 'localhost'),
            'port' => env('DB_PORT', '1433'),
            'database' => env('DB_DATABASE', 'forge'),
            'username' => env('DB_USERNAME', 'forge'),
            'password' => env('DB_PASSWORD', ''),
            'charset' => 'utf8',
            'prefix' => '',
            'prefix_indexes' => true,
        ],
    'redis' => [
```

```
        'client' => env('REDIS_CLIENT', 'phpredis'),

        'options' => [
            'cluster' => env('REDIS_CLUSTER', 'redis'),
            'prefix' => env('REDIS_PREFIX', Str::slug(env('APP_NAME', 'laravel'),
'').'_database'),
        ],

        'default' => [
            'url' => env('REDIS_URL'),
            'host' => env('REDIS_HOST', '127.0.0.1'),
            'password' => env('REDIS_PASSWORD'),
            'port' => env('REDIS_PORT', '6379'),
            'database' => env('REDIS_DB', '0'),
        ],

        'cache' => [
            'url' => env('REDIS_URL'),
            'host' => env('REDIS_HOST', '127.0.0.1'),
            'password' => env('REDIS_PASSWORD'),
            'port' => env('REDIS_PORT', '6379'),
            'database' => env('REDIS_CACHE_DB', '1'),
```

APPENDIX F

Plagiarism Result

Grammarly Results Chapter 1-3

Report: grammarly 1-3

grammarly 1-3

by Ronah joy Ramirez

General metrics

82,555	12,259	896	49 min 2 sec	1 hr 34 min
characters	words	sentences	reading time	speaking time

Score



This text scores better than 81%
of all texts checked by Grammarly

Writing Issues

440	171	269
Issues left	Critical	Advanced

Plagiarism

This text hasn't been checked for plagiarism

Grammarly Results Chapter 4-5

Report: grammarly 4-5

grammarly 4-5

by Ronah joy Ramirez

General metrics

26,261	3,815	479	15 min 15 sec	29 min 20 sec
characters	words	sentences	reading time	speaking time

Score



This text scores better than 87% of all texts checked by Grammarly

Writing Issues

98	56	42
Issues left	Critical	Advanced

Plagiarism

This text hasn't been checked for plagiarism

Turnitin Results

DIGITAL WORKOUT: DEVELOPMENT OF RESPONSIVE WEB-BASED FOR GYM MEMBERSHIP AND MONITORING

ORIGINALITY REPORT

14%	8%	3%	7%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS

PRIMARY SOURCES

1	Submitted to Laguna State Polytechnic University Student Paper	2%
2	ir.kdu.ac.lk Internet Source	1%
3	www.coursehero.com Internet Source	1%
4	ijecom.org Internet Source	1%
5	Submitted to University of Stellenbosch, South Africa Student Paper	1%
6	Orange, Laura Michelle. "Gym Membership Wellness Program as a Cost Benefit to Health Insurance Companies.", University of Nevada, Reno, 2017 Publication	1%
7	dl.ucsc.cmb.ac.lk Internet Source	<1%
8	ualberta.scholaris.ca Internet Source	<1%
9	www.ijcstjournal.org Internet Source	<1%
10	Submitted to Politecnico de Leiria Student Paper	<1%

11	Submitted to Zamboanga Peninsula Polytechnic State University Student Paper	<1%
12	Submitted to Occidental Mindoro State College Student Paper	<1%
13	ajomc.asianpubs.org Internet Source	<1%
14	Submitted to Ghana Technology University College Student Paper	<1%
15	Jerzy Tchórzewski, Wojciech Nabiałek, Artur Księżopolski. "Analysis of a web application supporting gym management in terms of control and systems theory", Studia Informatica. System and information technology, 2023 Publication	<1%
16	Submitted to Central Philippines State University - Main Campus Student Paper	<1%
17	Hao-Yu Tsai, Jen-Chun Chang, Shiuh-Jeng WANG. "Chapter 13 A New Scheme Modeling Gym Membership Transactions with NFT Systems", Springer Science and Business Media LLC, 2025 Publication	<1%
18	ijsrem.com Internet Source	<1%
19	jurnal.utpas.ac.id Internet Source	<1%
20	Submitted to Institute of Research & Postgraduate Studies, Universiti Kuala Lumpur	<1%

APPENDIX G

Documentation

Interview with the Gym Admin





Survey with the Gym Admin





Survey with the Gym Members/Customers





APPENDIX H

Publishable Paper (IEE Format)

DIGITAL WORKOUT: DEVELOPMENT OF RESPONSIVE WEB-BASED FOR GYM MEMBERSHIP AND MONITORING

Nica Ela Paraguya
College of Computer Studies
Laguna State Polytechnic University
Laguna, Philippines
nicaelaparaguya02@gmail.com

Rodelio Raquel Jr.
College of Computer Studies
Laguna State Polytechnic University
Laguna, Philippines
jonjonvillanueva027@gmail.com

Jhonrio Gonzales *College of Computer Studies*
Laguna State Polytechnic University
Laguna, Philippines
honriogonzales228@gmail.com

Abstract – The gym plays an important role in improving both physical and emotional health. However, individuals who work during weekdays often have limited time to visit a gym for inquiries, registration, and progress monitoring. This capstone project aims to develop a smooth, accessible, and timely web-based system for gym administrators, members, and prospective members. Providing accessible and accurate gym information is a primary focus of the study. The developed website offers convenient online registration, real-time monitoring of members' physical progress, accessible payment methods, and efficient membership management. The system can be accessed anytime and anywhere using mobile devices, ensuring ease of use and flexibility. Separate login modules were created for administrators and members, allowing administrators to manage and monitor member records, while members can track their workout progress, payment dues, and other relevant information in real time. Overall, the system enhances convenience, efficiency, and accessibility in gym management and membership monitoring.

Keywords: *Gym Management, Workout Monitoring, ISO 25010, TAM, Convenience, Accessibility.*

I. INTRODUCTION

Project Context

The Technology began to emerge as the twenty-first century progressed, because of the programs or systems created by technology most of the works are getting easier. Use of computers in information and communication has become second nature. This is one of the most significant technological advancements and one of the most beneficial innovations. Business and company As a result, management activities are required to maintain the records of all individuals, including trainers, trainees, and employees. Maintaining data on paper, however, is incredibly time consuming, thus a computerized system that addresses all issues are essential. The Gym Management System is a computerized version of the manual system, the technology provides consumers with unparalleled data security. The gym management system is an easy-to-use gym membership management system. It helps keep records of members membership plan and allows communication between the owner and members. The gym management system is feature-packed, helping in the management and growth of the club. Jeffrey's Gym, located on the 2nd floor of Kalaw Building, Lopez Avenue corner National Highway, Los Baños, Laguna, operates as a fitness center offering

various training and exercise services. Despite its growth and increasing number of clients, the gym continues to rely on manual processes in managing its daily operations. All transactions, including membership registration, package tracking, instructor assignments, and payment recording, are done through handwritten entries in physical logbooks. This traditional method has led to several operational challenges, such as data loss, inaccurate records, and time-consuming processes that hinder the delivery of quality service to its members. As the demand for efficient service grows, the need for an automated system has become increasingly apparent.

Project Objectives

The general objective of this capstone project is to develop a web-based gym membership management system for Jeffrey's Gym would improve business operation, enhance customer

growth and improvement are greatly aided by computers. Systems programs are examples of innovations that make it easier to store and secure essential data. Gyms have become an important part of human life, providing the best training and bodybuilding facilities in a community.

convenience, and financial control. Specifically, this project aims the following:

1. To create an accessible and responsive web-based platform that provides a seamless user experience across all devices (desktop, tablet, and mobile) without the need for native app installation.
2. To implement core functional modules for user accounts, membership management, payment processing, and progress monitoring.
3. To evaluate the acceptability of the developed system using ISO 25010 and Technology Acceptance Model (TAM).

II. RELATED LITERATURE

Gym Management System

The Gym Management System (GMS) is a modern digital solution aimed at automating and optimizing the operational processes of gyms. This paper explores the design, implementation, and benefits of the GMS, focusing on enhancing efficiency, accuracy, and overall user experience. By integrating modern web technologies, the GMS streamlines membership management, class scheduling, and payment processing, addressing the shortcomings of traditional manual systems. The system also offers advanced reporting and analytics, empowering gym owners to make data-driven decisions for better business management. (Chokhandre et. Al, 2025).

Various models have been applied to analyze price volatility. Khabibah et al. [3] employed ARCH/GARCH models to study big cayenne prices in Malang District, Indonesia, finding low volatility at both producer and consumer levels. Calvo [4] demonstrated that sudden stops increase relative price volatility, using ARCH techniques to analyze the wholesale-to-consumer price ratio. Supriya and Mamilla [5] highlighted the importance of incorporating seasonality into price predictions using GARCH models, improving the accuracy of forecasting market movements.

The use of IOTA's technology to create a decentralized, fee-free system for tracking gym attendance and entry could be beneficial and an advantage to the gym as well as its members. Removing the inefficiencies and security threats linked to traditional methods can create an easy, safe, and resistant solution to record gym members' check-ins and check-outs in real-time. The goal is to transform the gym administration process with an automatic and decentralized solution that improves operations efficiently and enhances user experience. This decentralized system for tracking gym attendance intends to eliminate many of the current problems and difficulties faced by gym operators and members by utilizing IOTA's Tangle technology to provide a more effective and safer alternative for managing gym attendance. (Phung L. 2025)

Fitness Center: An automated System for gym Notification With client Attendance

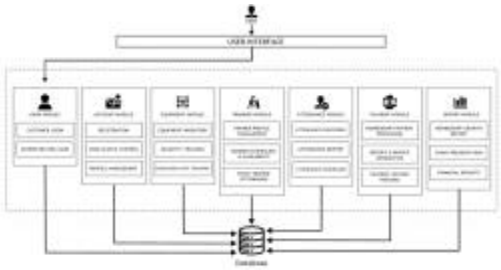
There are many people who are not able to use it due to various reasons. So, we are making an Android app which will help them. This project will allow the gym owner to manage all the receipts and also notify the users about their fees, mark their attendance by own, provide them proper diet plans as per the BMI and body type. Also this application will help them by providing exercise guidance clips.

Gyms and workout studios often only focus on getting people to purchase year-long memberships. It is very difficult for both the members and the trainer to keep all the paper receipts safe. Also it is difficult to manage all the client at a time and mark their attendance and provide exercises and diet plans. (Dadge R. et al. 2022)

Virtual gym management: Web based application to sell fitness products and equipment

In this paper our blueprint is to create a smooth web user interface and it can be run smoothly in windows or android devices. The web application will provide new updated calisthenics goods and users can buy using their credit or debit cards without hassle or hurdles in transaction. Gym Website, as described above, can lead to error free, secure, reliable and fast management system. It can assist the user to concentrate on their other activities products rather to concentrate on the record keeping. Thus it will help users to better utilization of resources in websites. The user credentials records saved in database without redundant entries. There will be a good communication between user and admin panel regarding goods quality and quantities and we will make sure to satisfy customers in every aspect. (P. N. Reddy et al. 2022)

This information allows for better planning, improved member engagement strategies, and effective operational management.



System Architecture

Project Development

Figure 6 illustrates the use of the Agile software development life cycle in designing the web-based system. The team adopted Agile Scrum, breaking the project into phases with 1–4-week iterations. This approach emphasized regular collaboration with users and continuous improvement. Key practices included managing a product backlog, sprint planning, daily scrums, and delivering a finished product at the end of each iteration.

III. METHODOLOGY

Project Design

This chapter applied architecture principles and practices to develop the system architecture that provides insight into the linkages between end-user requirements.

The system architecture shown in Figure 2 below includes a login feature that allows both gym members and staff (such as trainers and administrators) to set up accounts and enjoy a personalized fitness experience. he Account Module handles core user-related functions, including login, registration, profile management, and logout.

This module ensures secure authentication and role-based access for different types of users. It protects user information through proper validation and security measures.

The respondents and IT experts evaluated the developed project by using ISO 25010 software standard to study and learn about the performance and general features of the web-based system. The most popular way to determine the consistency of any program is ISO 25010. Assessments should include both functional and non-functional aspects, as well as internal and external design and structure.

Statistical Treatment

The study's population consists of key stakeholders involved in agricultural and poultry trading within the public markets of Cabuyao City, Bay, and Los Baños, all located in District 2 of Laguna. The primary participant groups are Vendors/Sellers who set prices, Consumers who are regular market buyers, and Market Administrators who manage market operations and pricing data. A purposive sampling technique employed to select an approximate total of 15 to 60 respondents. This targeted selection ensures a balanced representation from each stakeholder group to facilitate comprehensive data collection and gather essential user feedback for the system's development and subsequent evaluation.



Figure 6. Agile-Scrum Methodology

Evaluation Matrix

Standards, like ISO 25010 to ensure the quality of the developed system, these standards provide the important elements of a high-quality product. The ISO 25010 standards describe the quality model, which is the basis of a system for evaluating the quality of a product. The ISO 25010 contains eight quality attributes that describe assessment features of a system. The IT experts evaluated the developed project by the ISO 25010 software standard to study and learn about the performance and general features of the Web-based system. The most commonly used to determine the consistency of any program is ISO 25010.

Evaluation should consist of Functional and non-functional aspects, Internal and external design, and structure. To evaluate the system to be developed, the developers do the beta tests for the I.T experts. The developers used the ISO 25010 quality model as the standard for Likert scalar.

Ethical Consideration

To verify the protection of the data, this paper strictly apply ethical guidelines to assure the protection and rights of all the participants involved.

The protection of the participants through the application of ethical principles is significant in any research study. First communicate to the owner of Jeffry's gym to inform them and ask to permission in conducting interview. And provides a consent letter with the signature of our research adviser to be given to our client and for us to have a permission to conduct the interview and collect the data. To ensures that the participants should understand what they will be asked during the interview. This means, to participate in a capstone project, participants need to be informed about the research and it depends on them if they are willing to participate or not. Assure that the data we collect and the information that the client said during the interview is safe and won't be disclosed to other. The question used were checked by the research adviser before it was carried out.

IV. RESULT AND DISCUSSION

Table 21 presents the results of the acceptance testing of the system based on various quality characteristics. The findings indicate that the system achieved high mean scores across all evaluated criteria, reflecting a high level of user acceptance. Among the characteristics, Flexibility obtained the highest mean score (M = 4.69, SD = 0.04), followed by Maintainability (M = 4.65, SD = 0.22) and Security (M = 4.54, SD = 0.17), all of which were interpreted as Excellent. These results suggest that users strongly perceived the system as adaptable to changing requirements, easy to maintain, and secure in handling data and system operations also received Excellent interpretations, indicating that the system effectively meets user requirements, supports smooth user interaction, operates reliably, and ensures safe usage. Although Furthermore, while Performance Efficiency and Compatibility were rated as Good, their scores still signify that the system performs adequately and functions well across different platforms and environments. These results imply that the system is already effective in real-world use, with opportunities for future enhancements to further improve speed, resource utilization, and interoperability. In general, the findings confirm that the system is highly acceptable to users and meets the required quality standards, making it suitable for deployment and continued use. for improvement in performance efficiency and compatibility, these aspects do not significantly hinder overall system acceptance. Instead, they present opportunities for future enhancements that can further strengthen system performance and broaden its applicability across various platforms.

Performance Efficiency (M = 4.08, SD = 0.26) and Compatibility (M = 4.12, SD = 0.10) were interpreted as Good, their mean scores remain high, suggesting satisfactory.

Characteristics	Mean	SD	Interpretation
Functional Sustainability	4.27	0.24	Excellent
Performance Efficiency	4.08	0.26	Good
Compatibility	4.12	0.10	Good
Interaction Capability	4.38	0.20	Excellent
Readability	4.43	0.24	Excellent
Security	4.64	0.17	Excellent
Maintainability	4.65	0.22	Excellent
Flexibility	4.69	0.04	Excellent
Safety	4.45	0.08	Excellent
Overall Mean	**4.40**	0.08	**Excellent**

n=50

Overall, the acceptance testing results indicate that the system demonstrates a high level of quality and reliability across most evaluated characteristics. The Excellent ratings in Flexibility, Maintainability, and Security highlight the system's strong capability to adapt to changing user needs, support ease of maintenance, and protect data and system operations. These strengths reflect a well-structured and user-centered design that promotes long-term usability and trust among users.

The development and evaluation process followed industry standards to ensure that the system is user-friendly, functional, secure, and reliable. Based on feedback from users and evaluators, the Digital Workout PWA successfully demonstrates its ability to improve gym operations and support member engagement.

Taken together, the acceptance testing results validate the system's quality, effectiveness, and user acceptance, confirming that it meets established standards and is suitable for continued implementation and future development.

V. SUMMARY, CONCLUSION AND RECOMMENDATION

The study developed Digital Workout: A Progressive Web Application for Gym Membership and Monitoring to provide a modern and efficient solution for managing gym operations. Traditional gym processes—such as manual encoding of member records, attendance logging, payment tracking, and monitoring of workout progress—often result in delays, inaccuracies, and inconvenience for both gym staff and members. To address these issues, the researchers designed a PWA-based system that can be accessed on any device, offering flexibility, faster transactions, and real-time monitoring.

The system includes key features such as member registration, attendance tracking using digital logs, membership payment monitoring, and a dashboard for viewing member information and progress. Gym staff can easily update records, while members can view their personal data and monitor their activity anytime

The system allows staff to manage memberships and track payments efficiently, reducing paperwork and minimizing errors.

REFERENCES:

[1] Bakar N. & Said M. (2019). The Development of an Online Gym Booking System (OGBS) using Online Membership Business Model

[2] Chokhandre S. et. Al. (2025). Gym Management System

[3] Ebrahimi A. et Al. (2017). Gym-Suite an Extensible Platform for Gyms Management.
Delgado R. H. (2021). Gym-Suite An extensible platform for gyms management

[4] Gamage T. et. Al. (2022). Gym Management in Sri Lanka

[5 Hadawale S. et. Al. (2025). The Role of Web-Based Gym Management Systems in Enhancing User Experience

[6] Hambali M. H. et. Al. (2024). Design and Evaluation of a Web-based Membership Management System

[7] Hui L. & Teo P. (2024). An Implementation of Digital Platform to Enhance
the Appointment Scheduling System

[8] Jain A. (2015). A smart gym framework: Theoretical approach
[9] Karmakar A. (2022). Gym Management System

[10] Khattri V. et. Al. (2024). Online Payment Transaction System
[11] Kendiga J. (2017). Taurama Fitness Centre System
[12] Kotzé M. A. (2017). A framework for the steps towards a smart gym

[13] Kumar Dinesh A.V. et. Al. (2020). Smart Gym Management System

[14] Lakshitha K. & Sirisuriya S. (2021). An Automated Platform to Manage Customer Relationship in a Gymnasium

[15] Löthman & Liljeblad. (2021). Committed to Exercise: A qualitative study on the persuasive effect and potential of a gym's online booking system

[16] Monir A. & Jannatun N. (2016). A Project Paper on Smart Gym Management

[17] Nivedita V. et. Al (2025). Web-based fitness trainer and client's management system

[18] Orange L. M. (2017). Gym Membership Wellness Program as a Cost Benefit to Health Insurance Companies

[19] Zhao D. et al. (2023). Design and Implementation of Gym Management System

[20] Vazhachariccka P. J. (2020). Gym management interface

[21] Vaishnavi Reddy G. et. Al. (2024). Digital Gym Management Systems: User Experience & Efficiency Study

APPENDIX I

Curriculum Vitae

NICA ELA D. PARAGUYA

0907 295 2792
nicaelaparaguya02@gmail.com
Purok 6, Brgy 2 Tibag Calamba Laguna



PERSONAL DETAILS

Date of Birth: December 29, 2000
Birth Place: Sta Cruz, Manila
Civil Status: Single
Sex: Female
Height: 5'3
Weight: 48kg
Nationality: Filipino
Religion: Christian

SKILL

- Microsoft Office (Word, Excel, PowerPoint)
- Excellent Communication
- Basic HTML & CCS
- Basic Graphic Design
(Photoshop, Figma, Canva)
- Basic Video Editing
(Adobe Premiere Pro, CapCut)

SEMINARS

- -Securing the Future: Understanding the Fundamentals of Cyber Security – December 12, 2023
- -Role of Technology in Sustainable Development – February 28, 2025

EDUCATION

Tertiary
LAGUNA STATE POLYTECHNIC UNIVERSITY
Bgry. Mayondon-Malinta, Los Banos, Laguna
2022-Present
Bachelor of Science in Information Technology – Major in Service Management Program (SMP)
2022 - Present

Secondary – Senior High School
AMA University of Calamba
Information and Communications Technology (ICT) Strand
National Hi-way Parian, Calamba City, Laguna
2017- 2019

Junior High School
Makiling Integrated National High School
National Highway, Brgy. Makiling, Calamba City, Laguna
2013 - 2017

Elementary School
Apolinario Mabini Elementary School
Severino St, Quiapo, Manila
2007 - 2012

IBM SKILLBUILD DIGITAL CREDENTIALS

- AI Fundamentals: Foundations for Understanding AI
- Cybersecurity Fundamentals
- Web Development Fundamentals
- Information Technology Fundamentals
- Working in a Digital World: Professional Skills

RODELIO V. RAQUEL JR.

09095220312
jonjonvillanueva027@gmail.com
#53 Upper Dampalit Lalakay Los Baños Laguna



PERSONAL DETAILS

Date of Birth: October 07, 2004
Birth Place: Los Baños, Laguna
Civil Status: Single
Sex: Male
Height: 5'5
Weight: 60kg
Nationality: Filipino
Religion: Christian

SKILL

- Microsoft Office (Word, Excel, PowerPoint)
- Basic HTML & CCS

SEMINARS

- ·Securing the Future: Understanding the Fundamentals of Cyber Security – December 12, 2023
- ·Role of Technology in Sustainable Development – February 28, 2025

EDUCATION

Tertiary
LAGUNA STATE POLYTECHNIC UNIVERSITY
Bgry. Mayondon-Malinta, Los Banos, Laguna
2022-Present
Bachelor of Science in Information Technology – Major in Service Management Program (SMP)
2022 - Present

Secondary – Senior High School
Lalakay Senior High School
Technical Vocational Livelihood(TVL) Strand
Brgy. Lalakay Los Baños ,Laguna
2020-2022

Junior High School
Los Baños National High School (LBNHS) Batong Malake
Brgy. Batong Malake Los Baños, Laguna
2016-2020

Elementary School
Lalakay Elementary School
Brgy. Lalakay Los Baños, Laguna
2009-2016

IBM SKILLBUILD DIGITAL CREDENTIALS

- Project Management Fundamentals
- Information Technology Fundamentals
- Web Development Fundamentals
- Cybersecurity Fundamentals

JHON RIO GONZALES

09213602890
jhonriogonzales228@gmail.com
Libis Lalakay, Los Baños, Laguna



PERSONAL DETAILS

Date of Birth: September 14, 2003
Birth Place: Calamba, Laguna
Civil Status: Single
Sex: Male
Height: 5'2
Weight: 47kg
Nationality: Filipino
Religion: Catholic

SKILL

- Microsoft Office (Word, Excel, PowerPoint)
- Basic HTML & CCS

SEMINARS

- ·Securing the Future: Understanding the Fundamentals of Cyber Security – December 12, 2023
- ·Role of Technology in Sustainable Development – February 28, 2025

EDUCATION

Tertiary
LAGUNA STATE POLYTECHNIC UNIVERSITY
Bgry. Mayondon-Malinta, Los Banos, Laguna
2022-Present
Bachelor of Science in Information Technology – Major in Service Management Program (SMP)
2022 - Present

Secondary – Senior High School
STI College Calamba
Brgy. Uno, Calamba City, Laguna
2019- 2021

Junior High School
Castor Alviar National High School
Brgy. Masili, Calamba City, Laguna
2015 - 2019

Elementary School
Tiyani Elementary School
Brgy. Sucol, Calamba City, Laguna
2009 - 2015

IBM SKILLBUILD DIGITAL CREDENTIALS

- Exploring Emerging Techs
- AI Fundamentals: Foundations for Understanding AI
- Data Fundamentals